

Greenhouse

Global Warming Science, EPS101

Camille Hankel and Eli Tziperman

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Class objectives

- How natural greenhouse gases keep the Earth from freezing

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- The logarithmic dependence of warming on CO₂
- How important are other greenhouse gasses relative to CO₂

Radiative forcing (RF)

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Observed radiative forcing increases

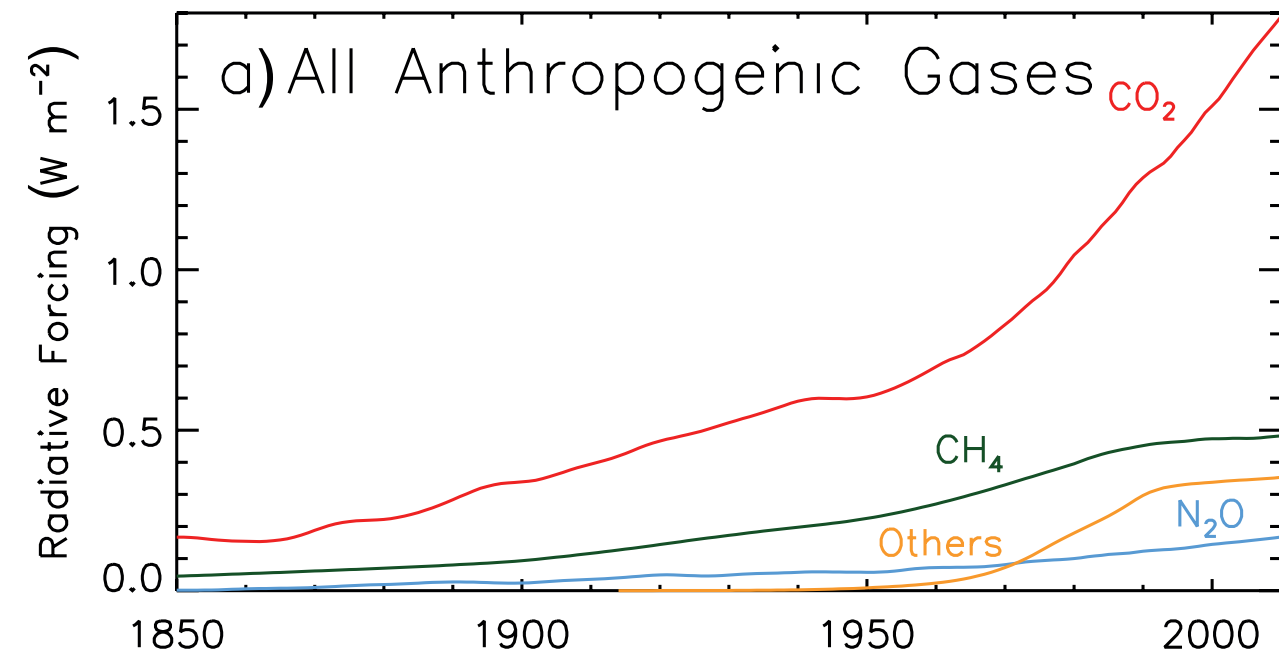


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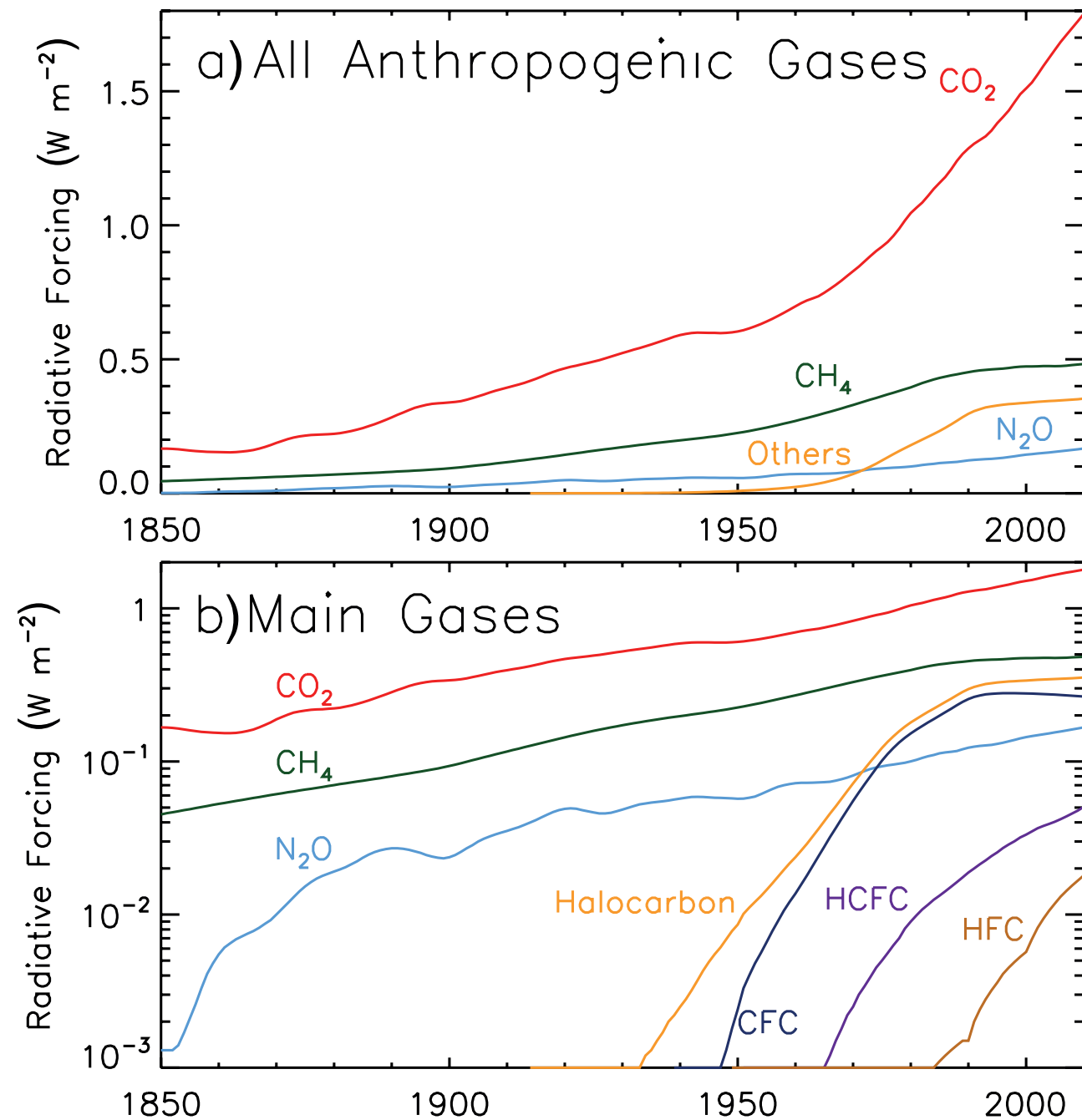


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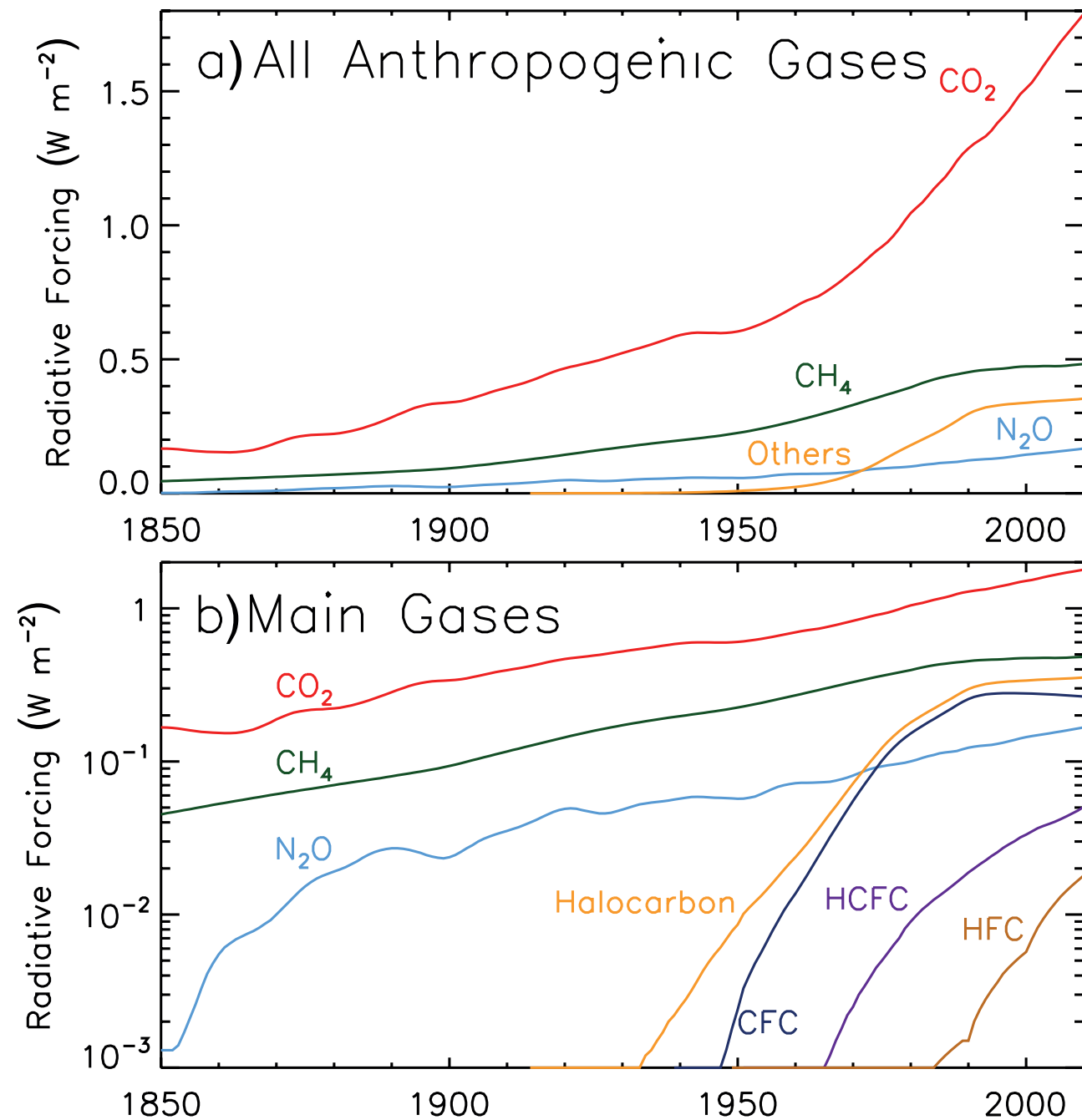
More specifically: (IPCC AR5, 2013)

RF: the change in net downward radiative flux at the tropopause after allowing for stratospheric temperatures to readjust to radiative equilibrium, while holding surface and tropospheric temperatures, and water vapor and cloud cover, fixed at the unperturbed values.

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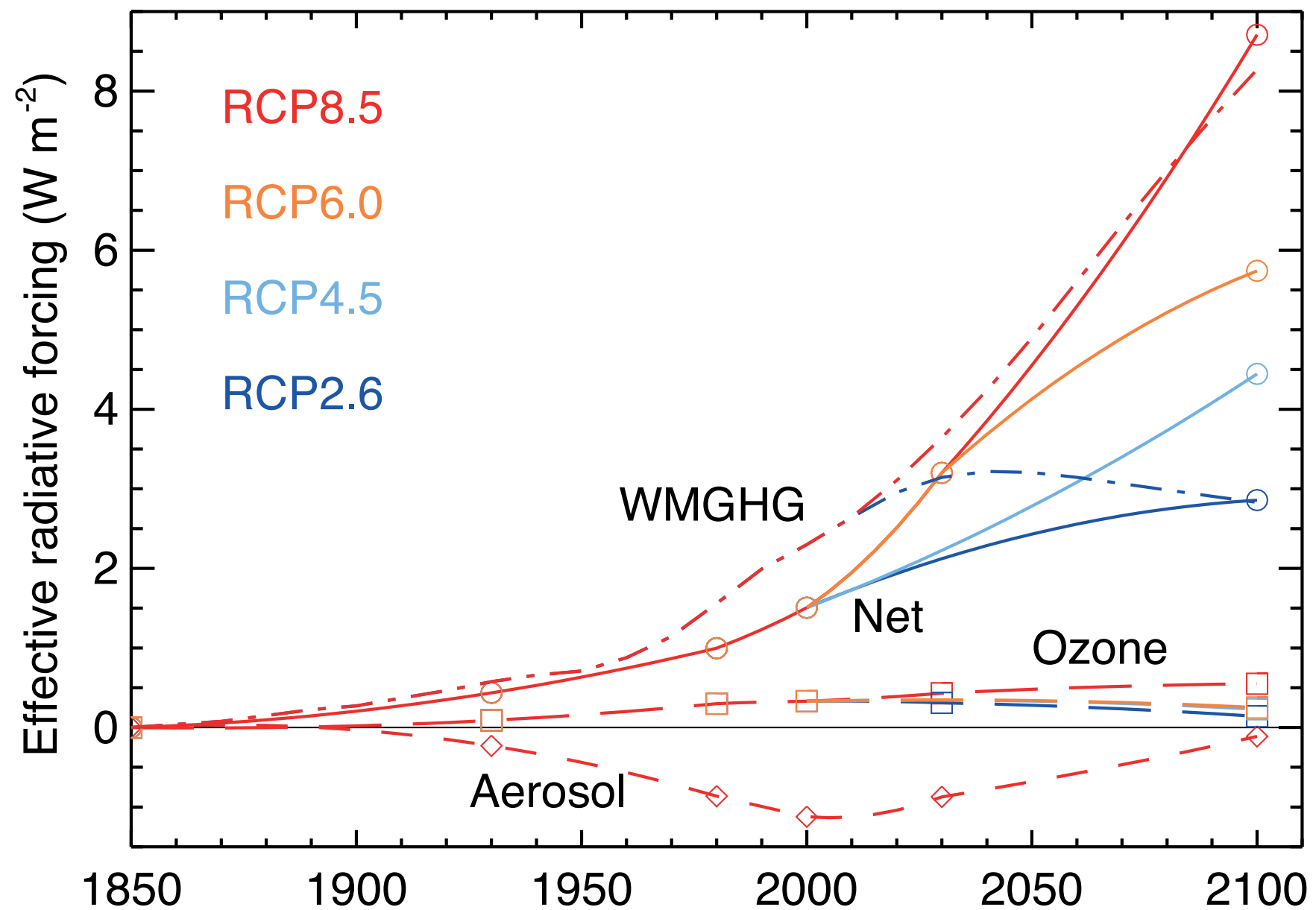


(IPCC AR5, 2013)

Future scenarios for radiative forcing: The Representative Concentration Pathway (RCP)

Future scenarios.

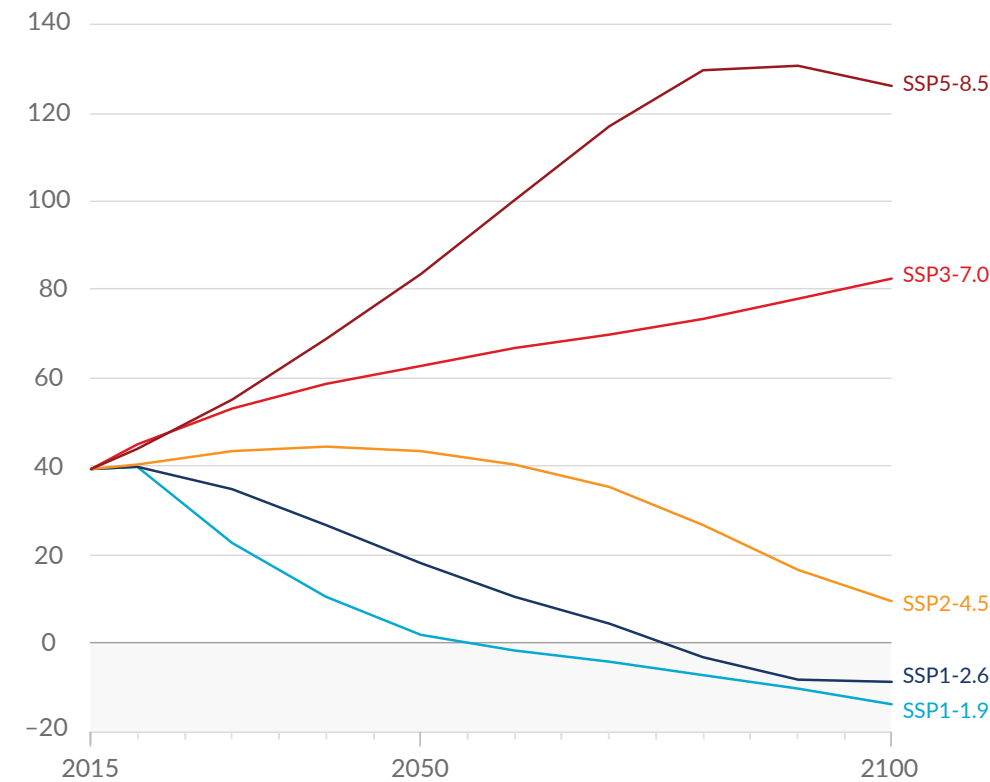
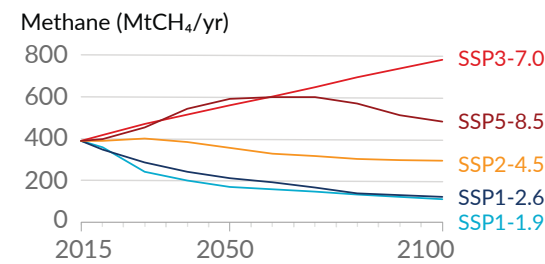
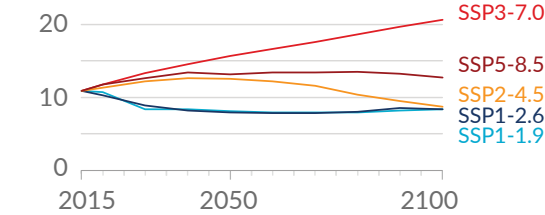
RCP8.5: worst-case scenario/ business as usual



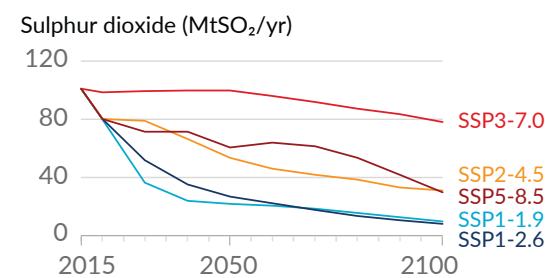
(IPCC AR5, 2013)

Figure 8.22 | Global mean anthropogenic forcing. (solid lines with circles are net; long dashes with squares are ozone; short dashes with diamonds are aerosol; dash-dot are WMGHG; colors: RCPs with **red for RCP8.5**, **orange RCP6.0**, **light blue RCP4.5**, and **dark blue RCP2.6**). Symbols: times of ACCMIP simulations were performed.

Shared Socioeconomic Pathways (SSPs)

Carbon dioxide (GtCO₂/yr)Selected contributors to non-CO₂ GHGsNitrous oxide (MtN₂O/yr)

One air pollutant and contributor to aerosols

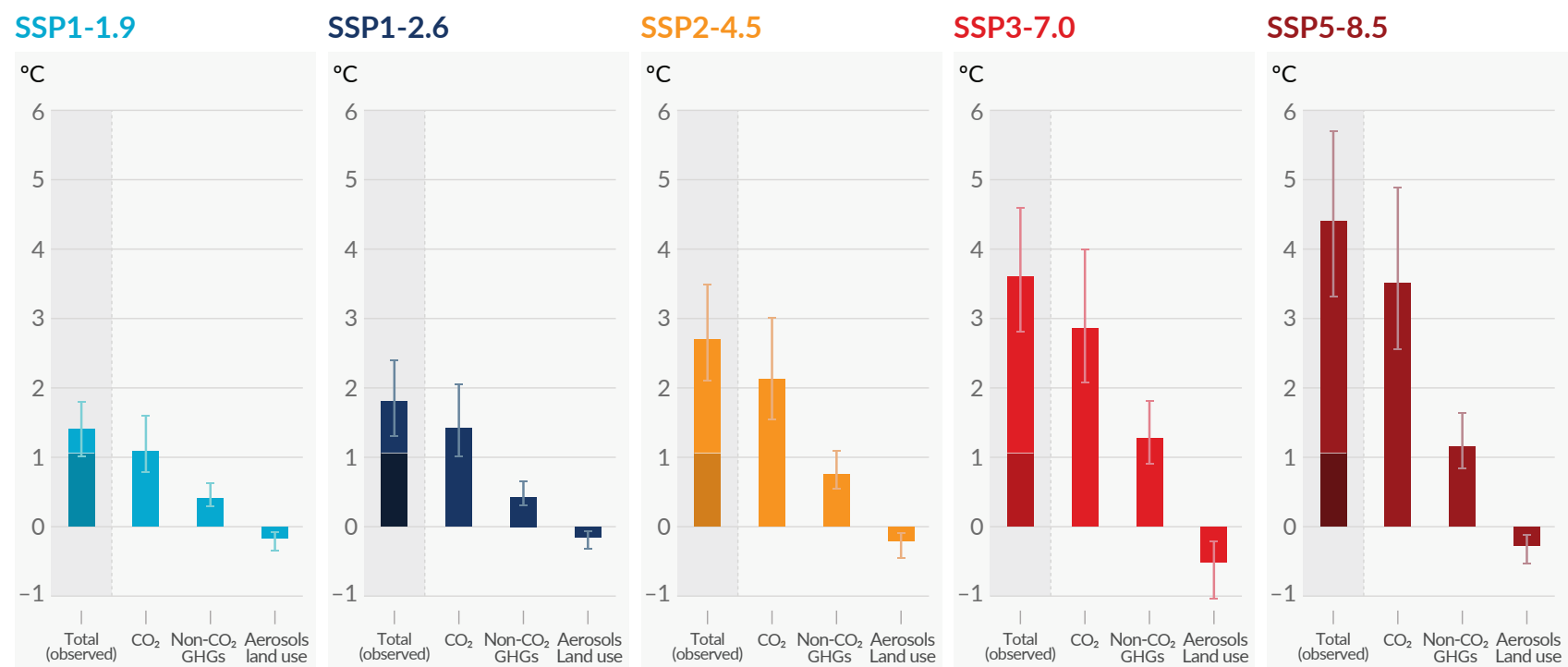


(a) Future annual emissions of CO₂ (left) and of a subset of key non-CO₂ drivers (right), across five illustrative scenarios

These are **emission** rather than **concentration** scenarios.

(b) Contribution to global surface temperature increase from different emissions, with a dominant role for CO₂ emissions

Change in global surface temperature in 2081–2100 relative to 1850–1900 (°C)



Total warming (observed warming to date in darker shade), warming from CO₂, warming from non-CO₂ GHGs and cooling from changes in aerosols and land use

(IPCC AR6, 2022)

Workshop #1:

Observed and projected increase in greenhouse gasses

Observations & projections of CO₂ concentration

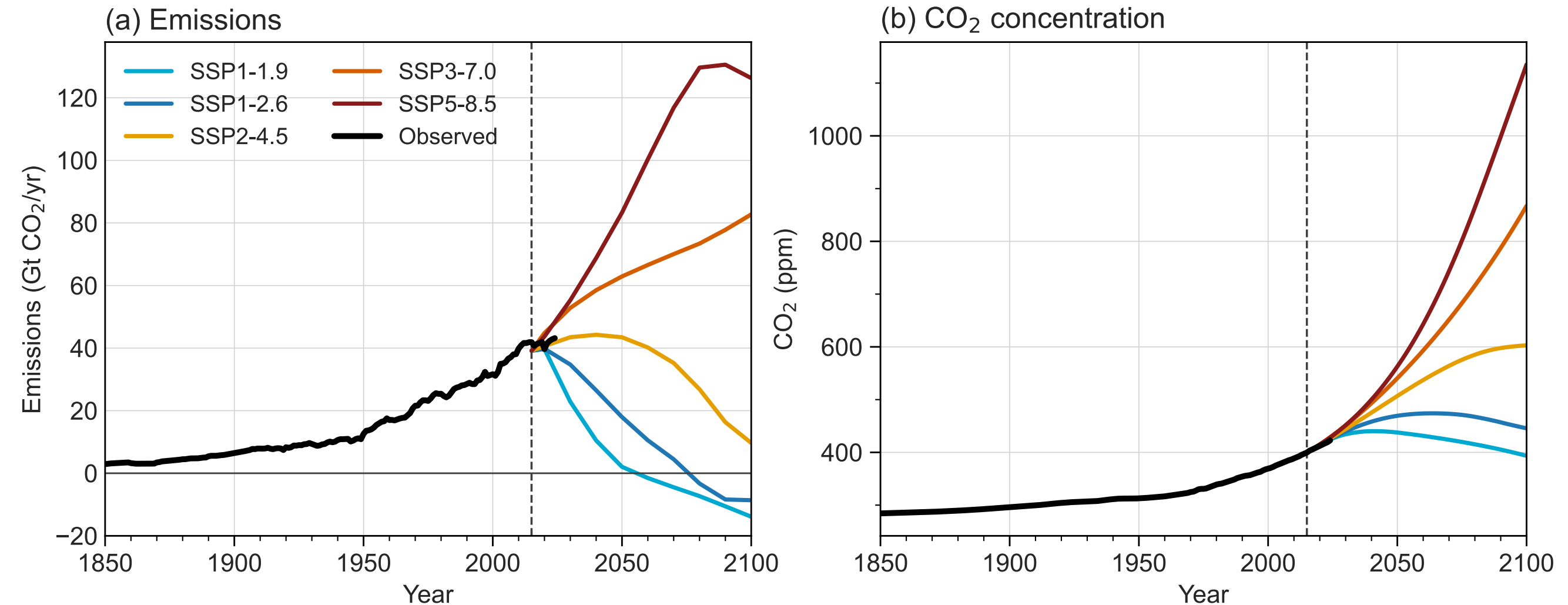
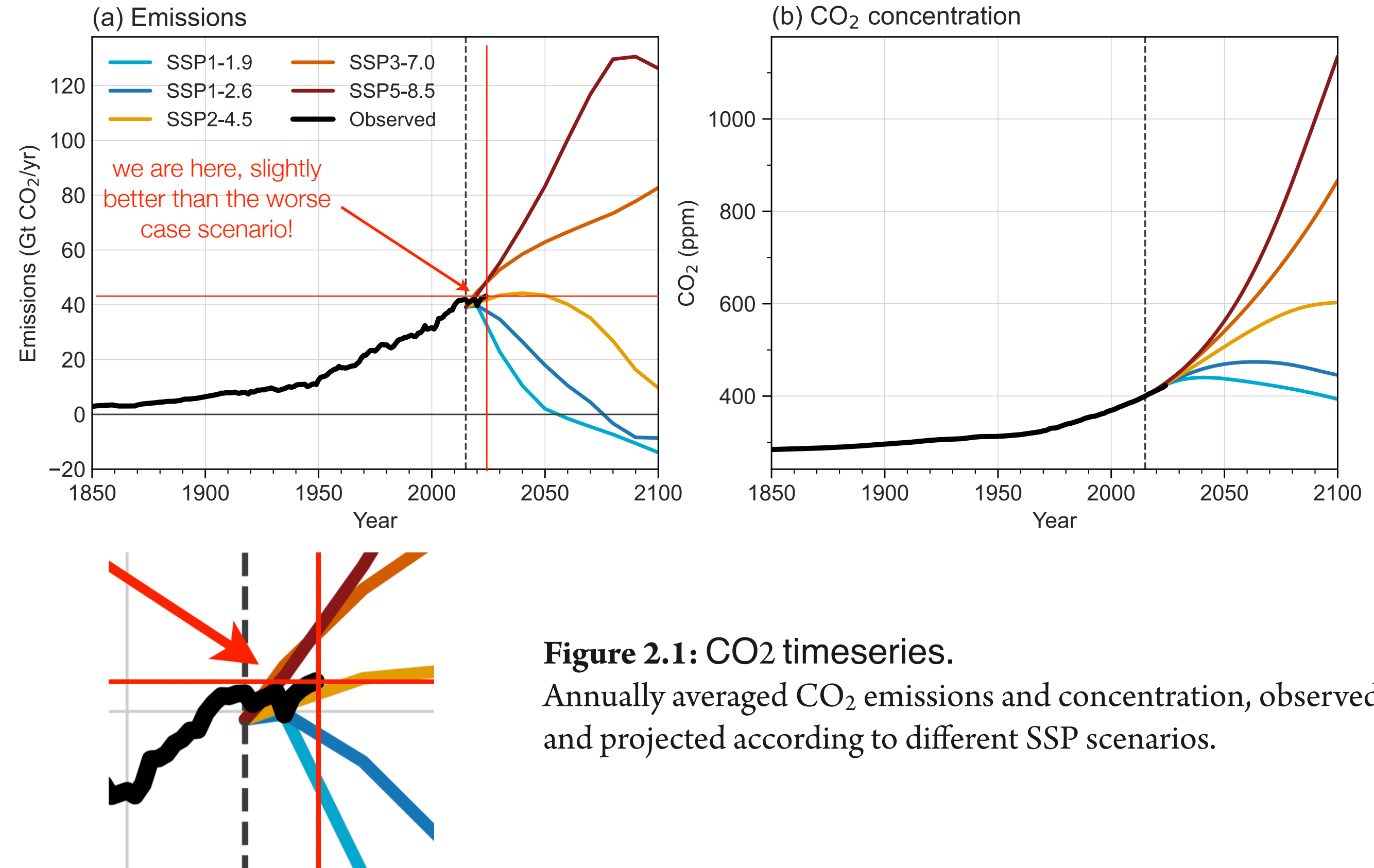


Figure 2.1: CO₂ timeseries.

Annually averaged CO₂ emissions and concentration, observed and projected according to different SSP scenarios.

Observations & projections of CO₂ concentration



Energy balance: Albedo, greenhouse

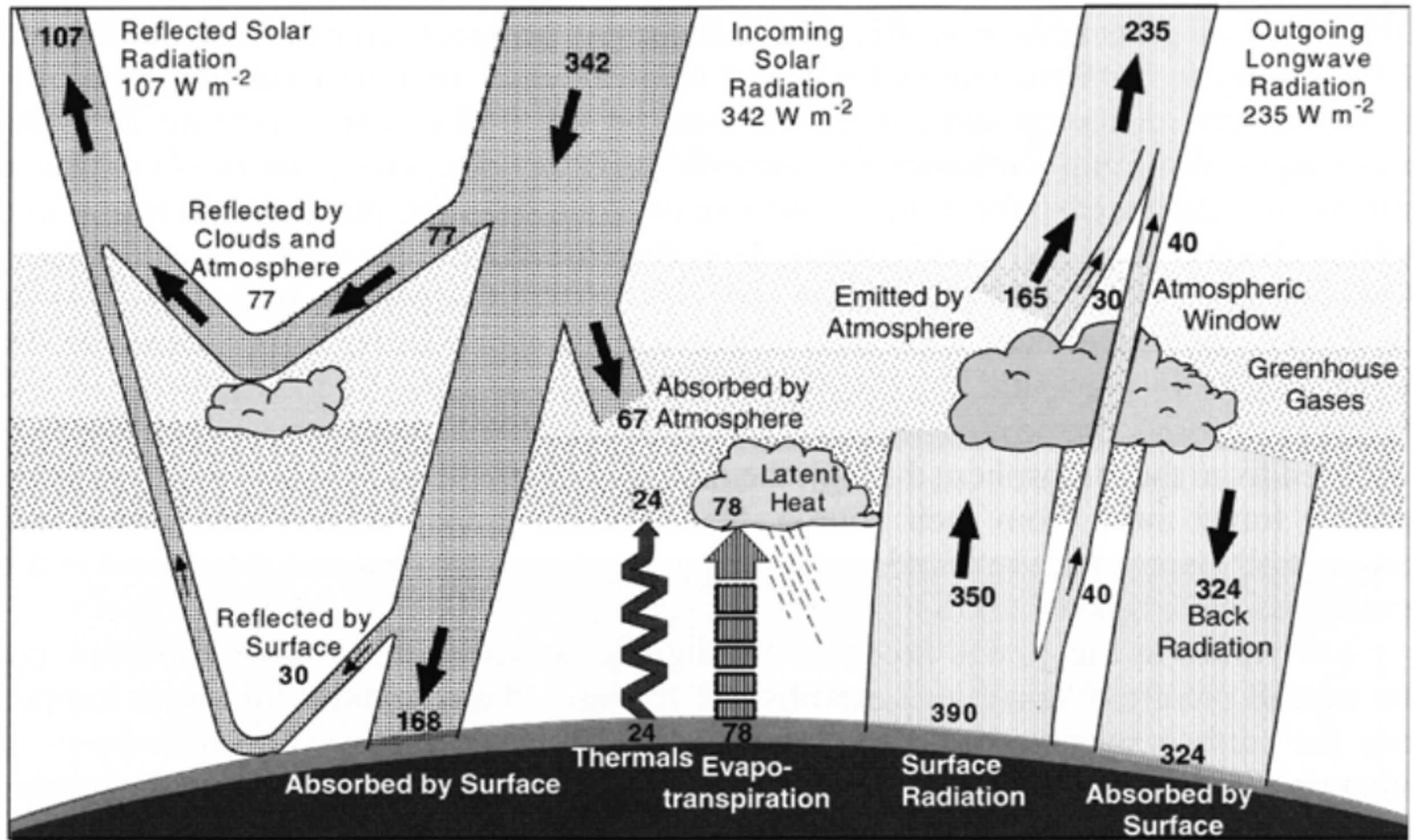


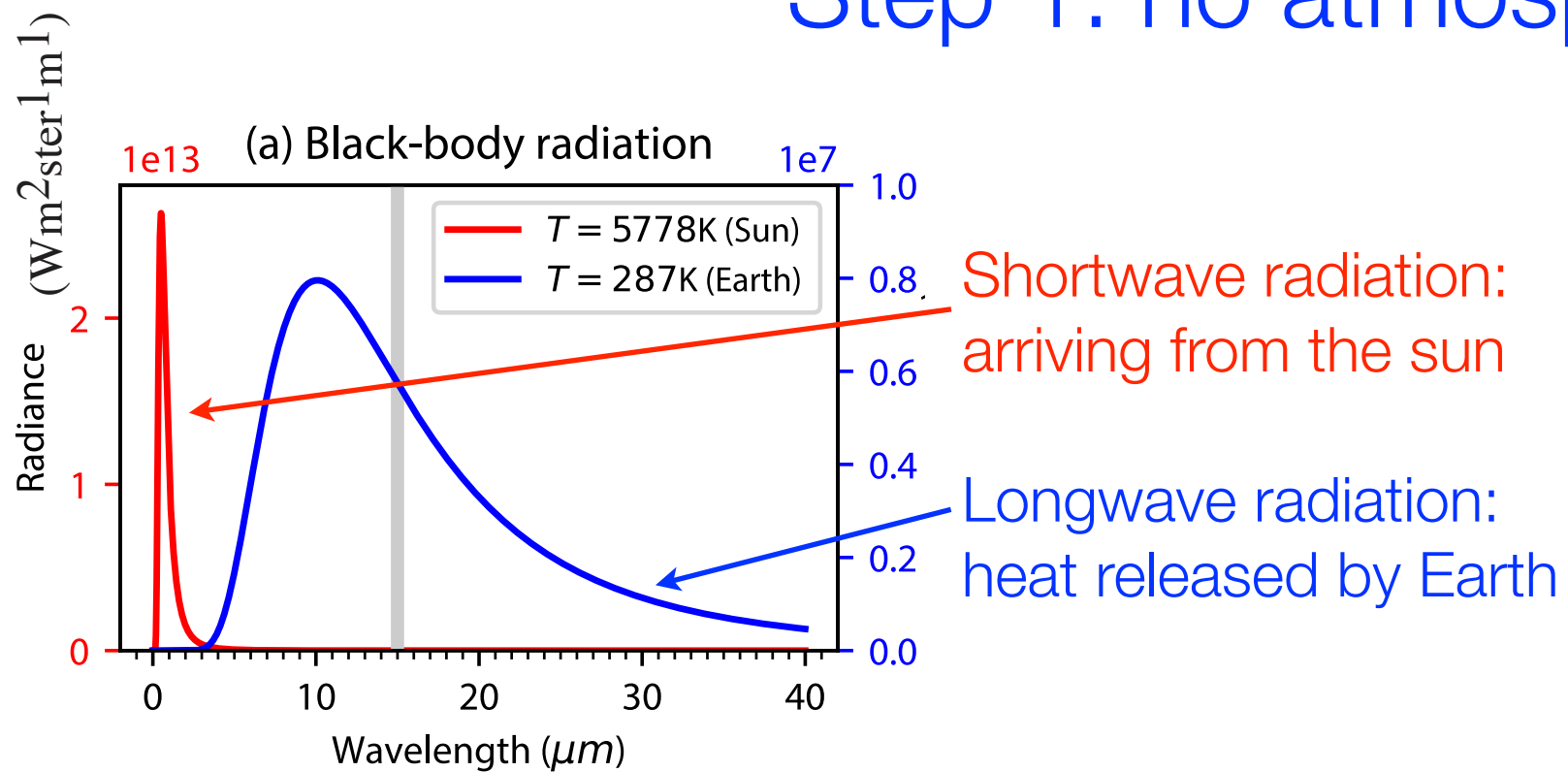
FIGURE 9.4. Earth's energy balance (from Trenberth, K.E., and D.P. Stepaniak, 2004: The flow of energy through the Earth's climate system. *Q.J.R.Meteorol.Soc.*, **130**, 2677-2701).

Notes sections 2.1.1, 2.1.2, 2.1.3:
Energy balance, 2-layer model, continuous temperature
profile and level of last absorption

(use next three slides)

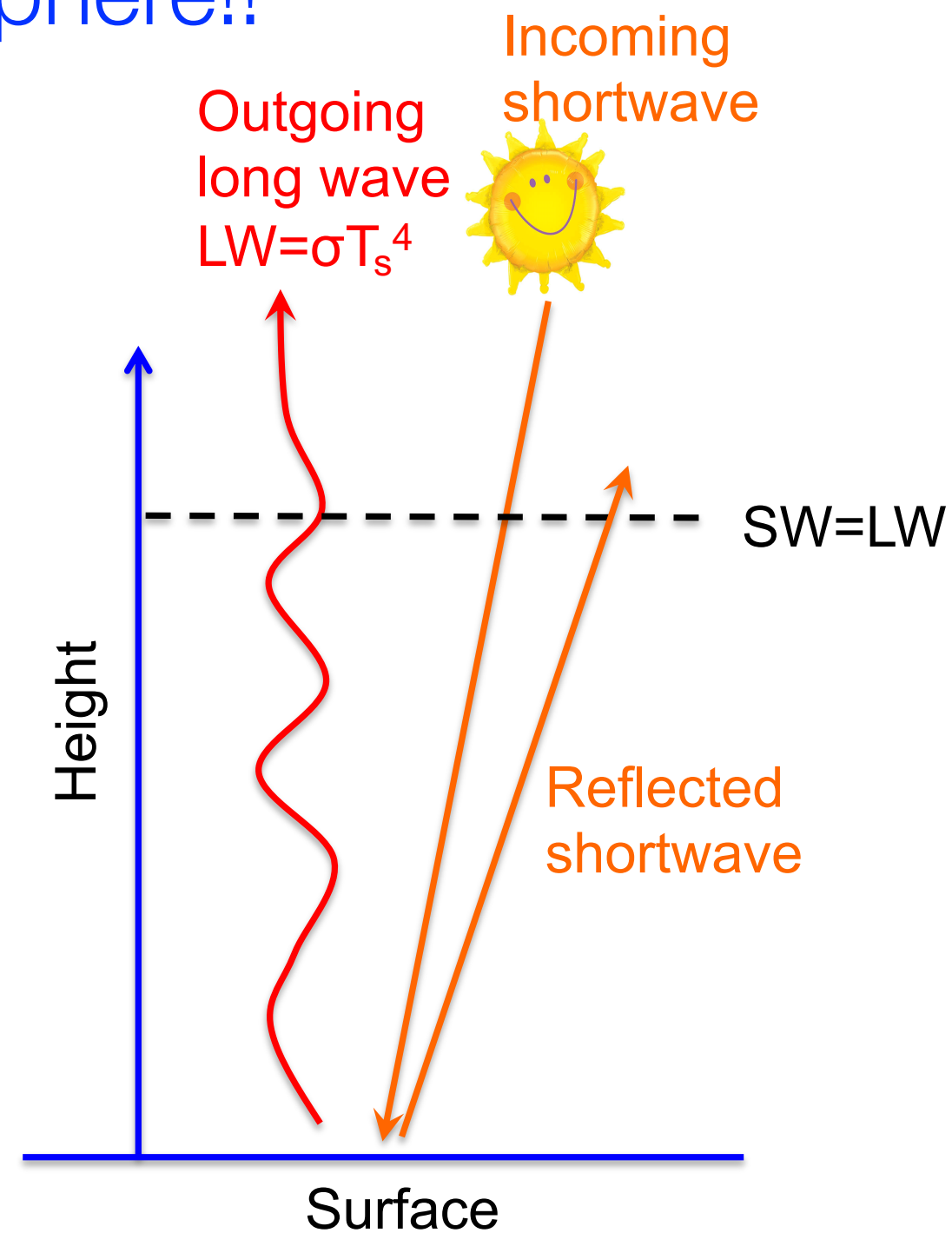
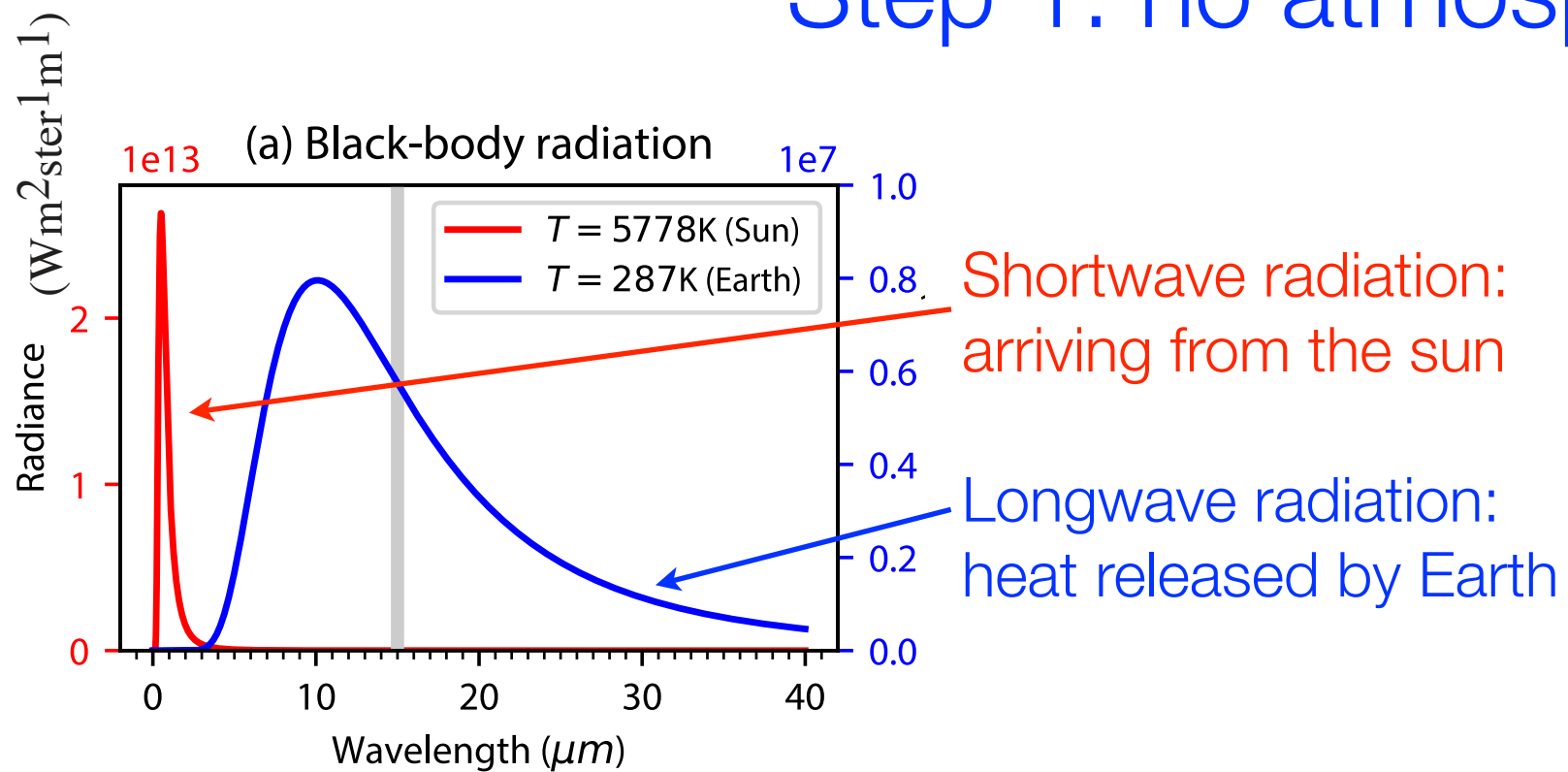
Energy balance of the Earth

Step 1: no atmosphere!!



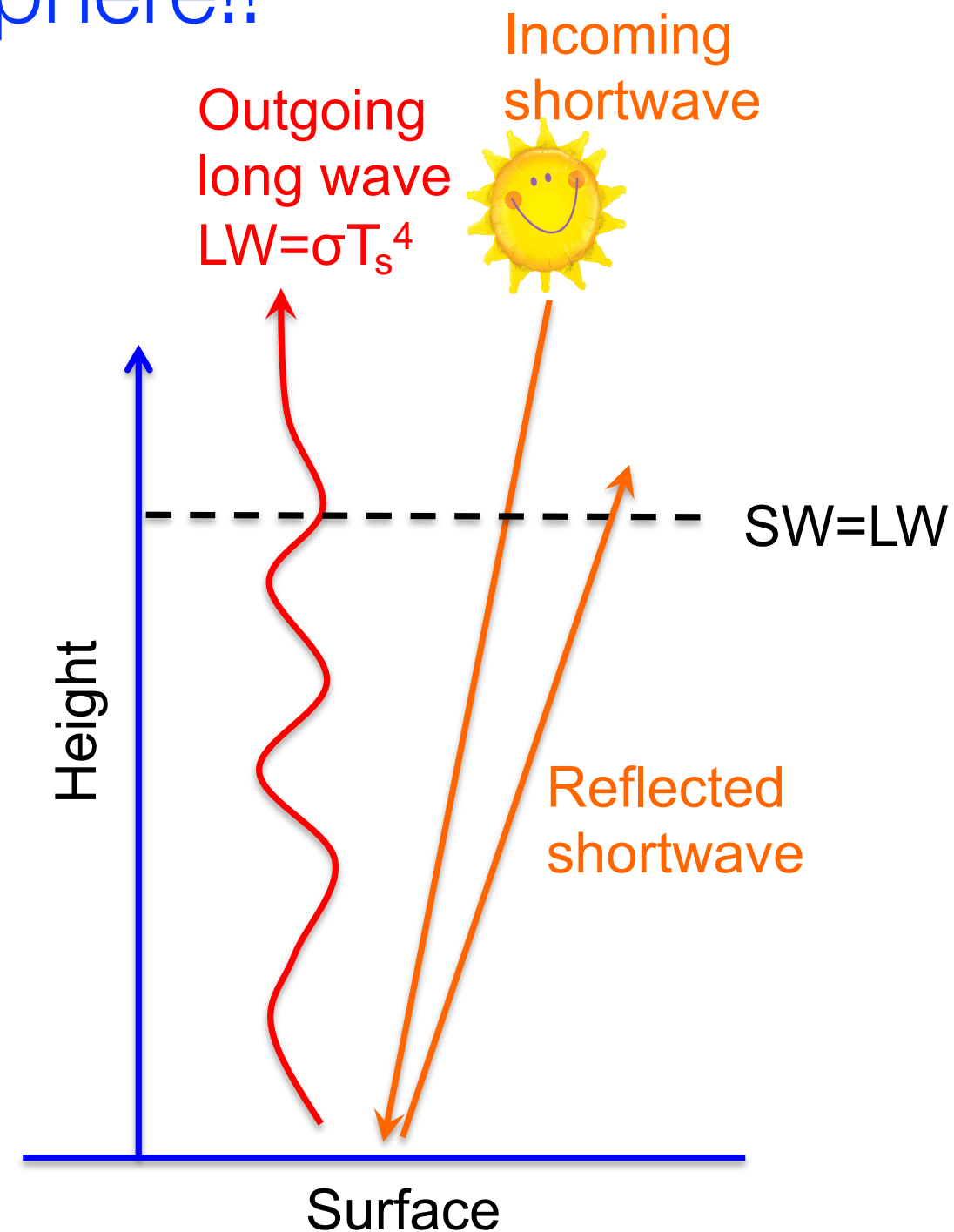
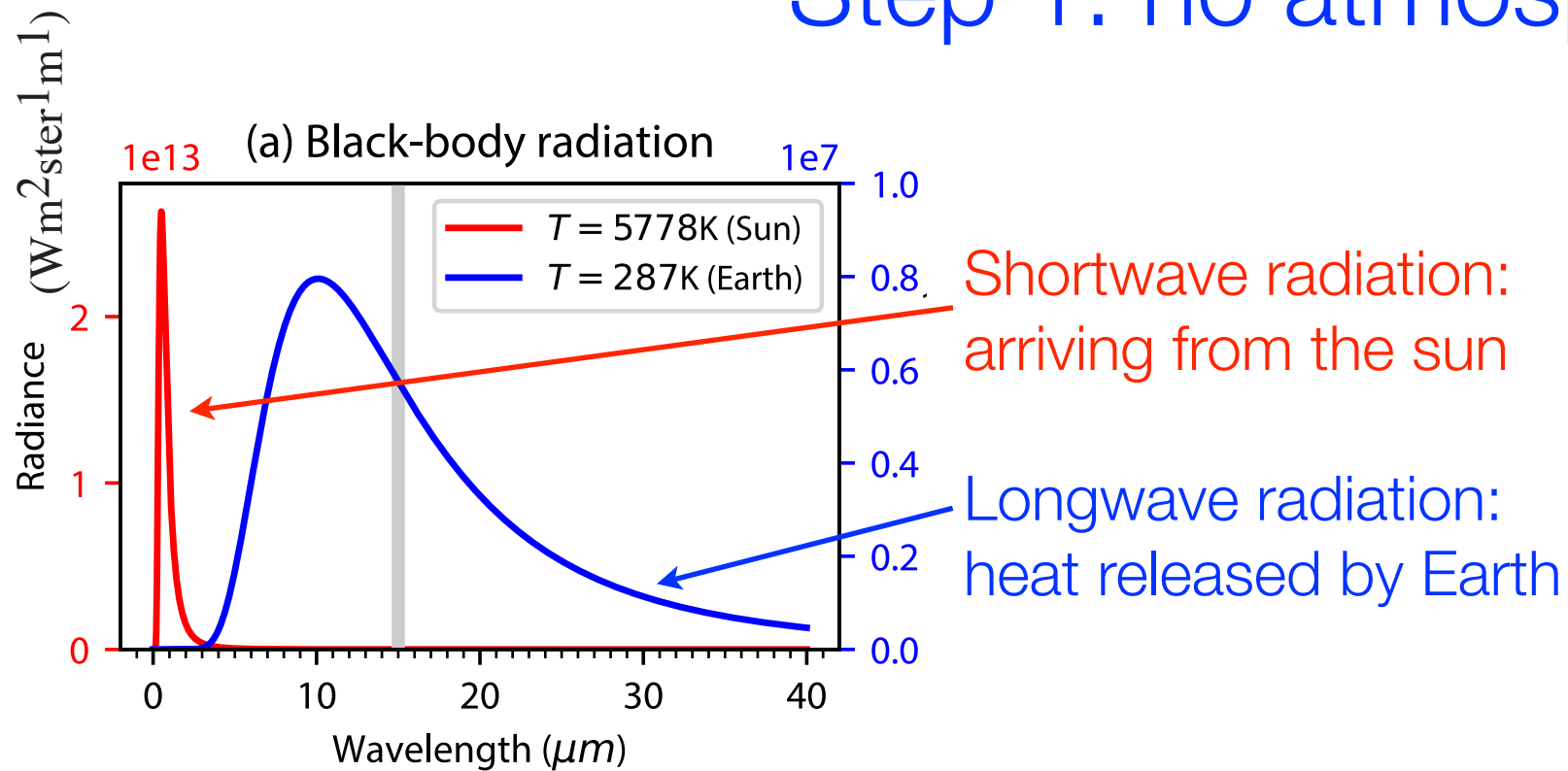
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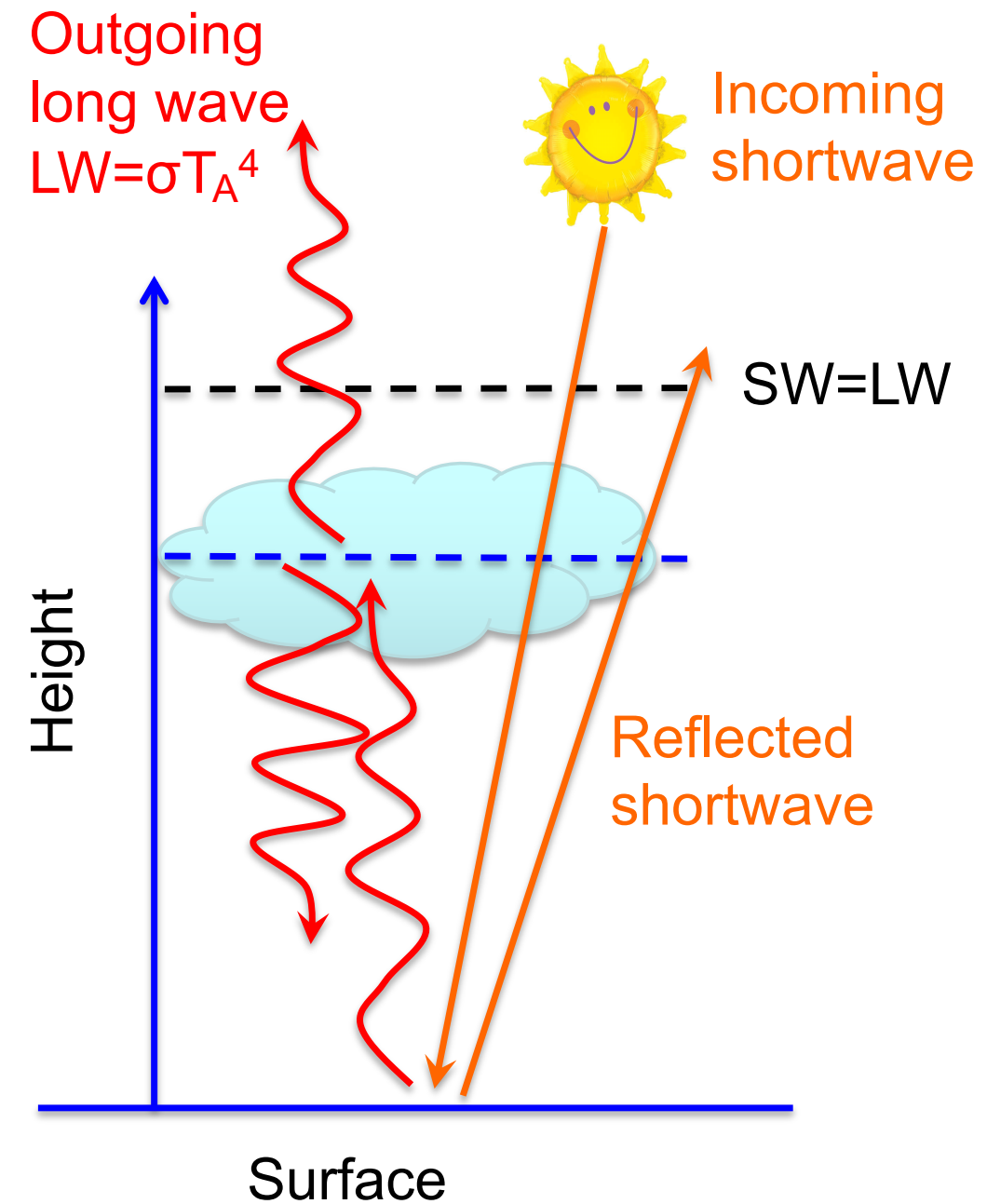
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- Energy conservation: incoming **SW** radiation to Earth = outgoing **LW** radiation to space
- Incoming **SW** = $\frac{S_o}{4}(1 - \alpha)$
- α = albedo = proportion SW reflected
- Outgoing **LW** = σT^4
- Set incoming = outgoing $\Rightarrow T = \left(\frac{(S_o/4)(1 - \alpha)}{\sigma} \right)^{1/4} = 255 \text{ K} = -18^\circ \text{C} \equiv T_0.$

The Greenhouse Effect

Step 2: add a 1-layer atmosphere



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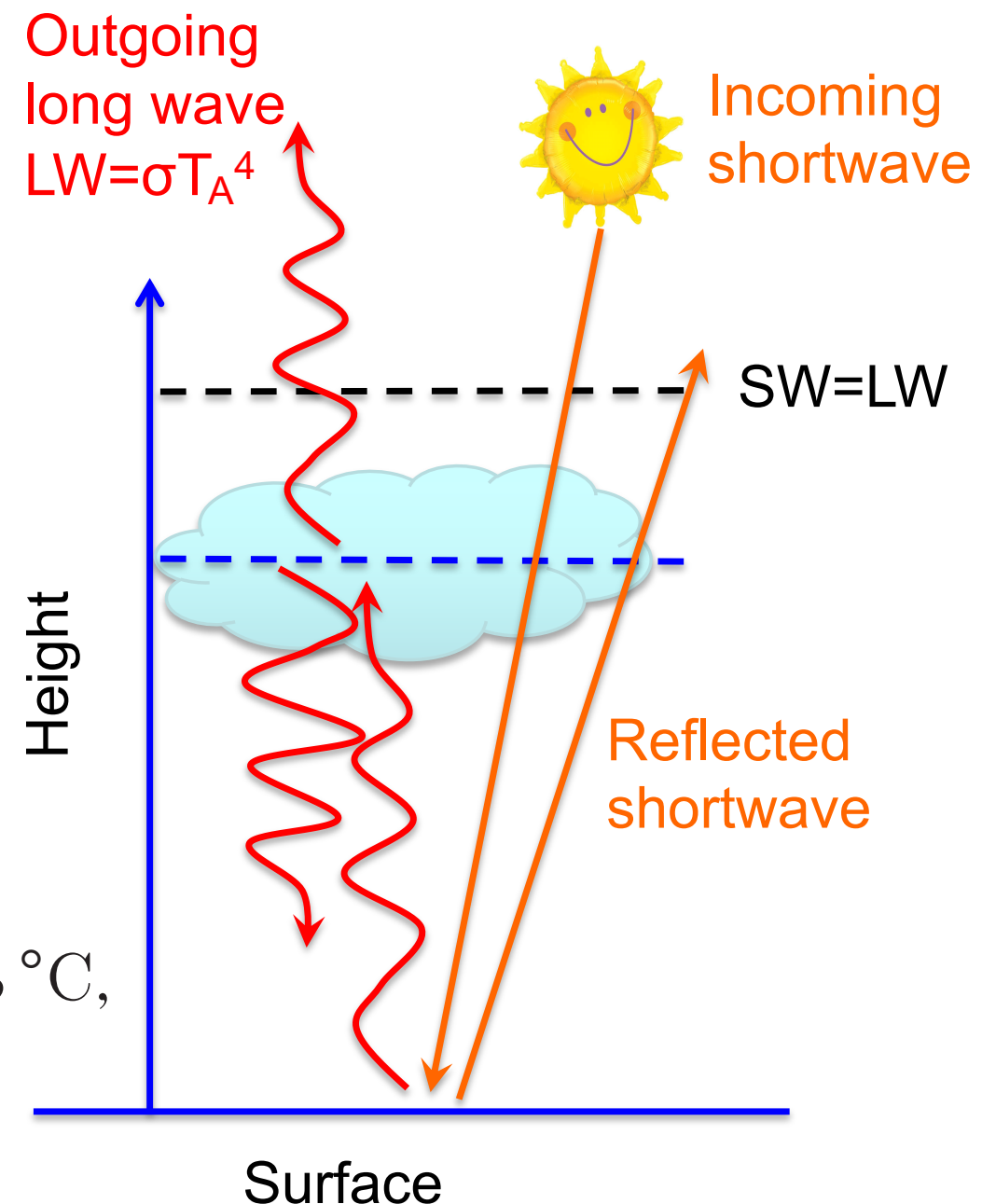
- Add an atmospheric layer which absorbs and re-emits LW radiation, transparent to SW
- Two unknowns: surface temperature T and (mid) atmospheric temperature T_a . Two equations (energy balance at the surface and at mid-atmosphere):

$$\frac{S_0}{4}(1 - \alpha) + \epsilon\sigma T_a^4 = \sigma T^4$$

$$\epsilon\sigma T^4 = 2\epsilon\sigma T_a^4.$$

$$T = \left(\frac{(S_0/4)(1 - \alpha)}{\sigma(1 - \epsilon/2)} \right)^{1/4} = T_0(1 - \epsilon/2)^{-1/4} = 284 \text{ K} = 13^\circ \text{C},$$

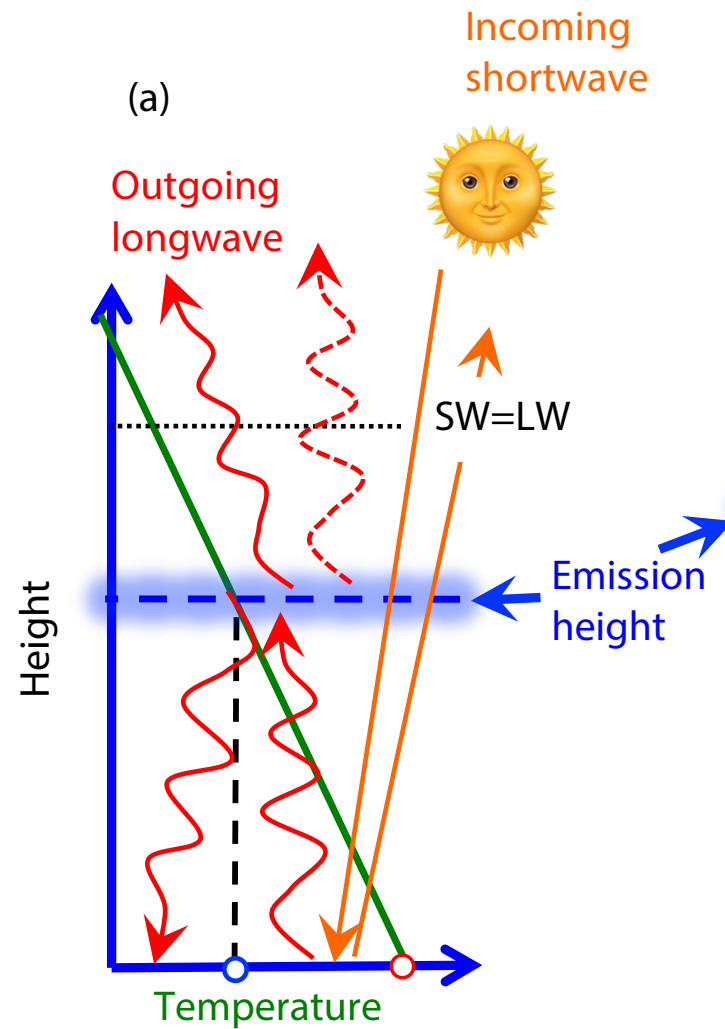
- Result: surface temperature is warmer due to the (natural) “greenhouse effect”.



Workshop #2 a,b: 2-layer energy balance model

The Anthropogenic Greenhouse Effect

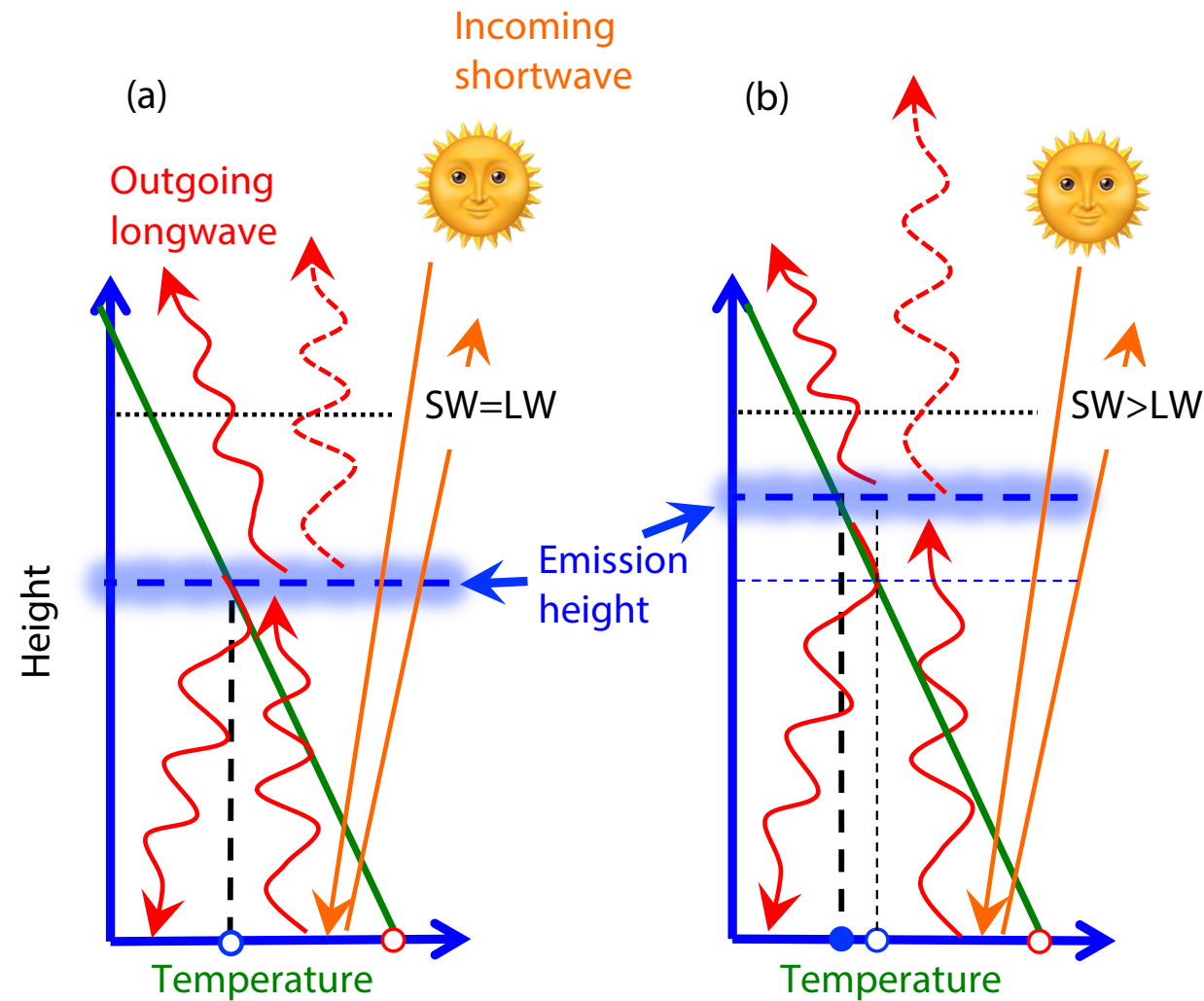
Step 3: add a continuous atmospheric temperature profile



- Level of last absorption: where most of the radiation emitted upward escapes to space, without getting absorbed again
- Increasing greenhouse gas ➡ Rising level of last absorption ➡ Earth radiates from a colder temperature ➡ Energy balance is broken: $LW < SW$ ➡ The temperature must adjust

The Anthropogenic Greenhouse Effect

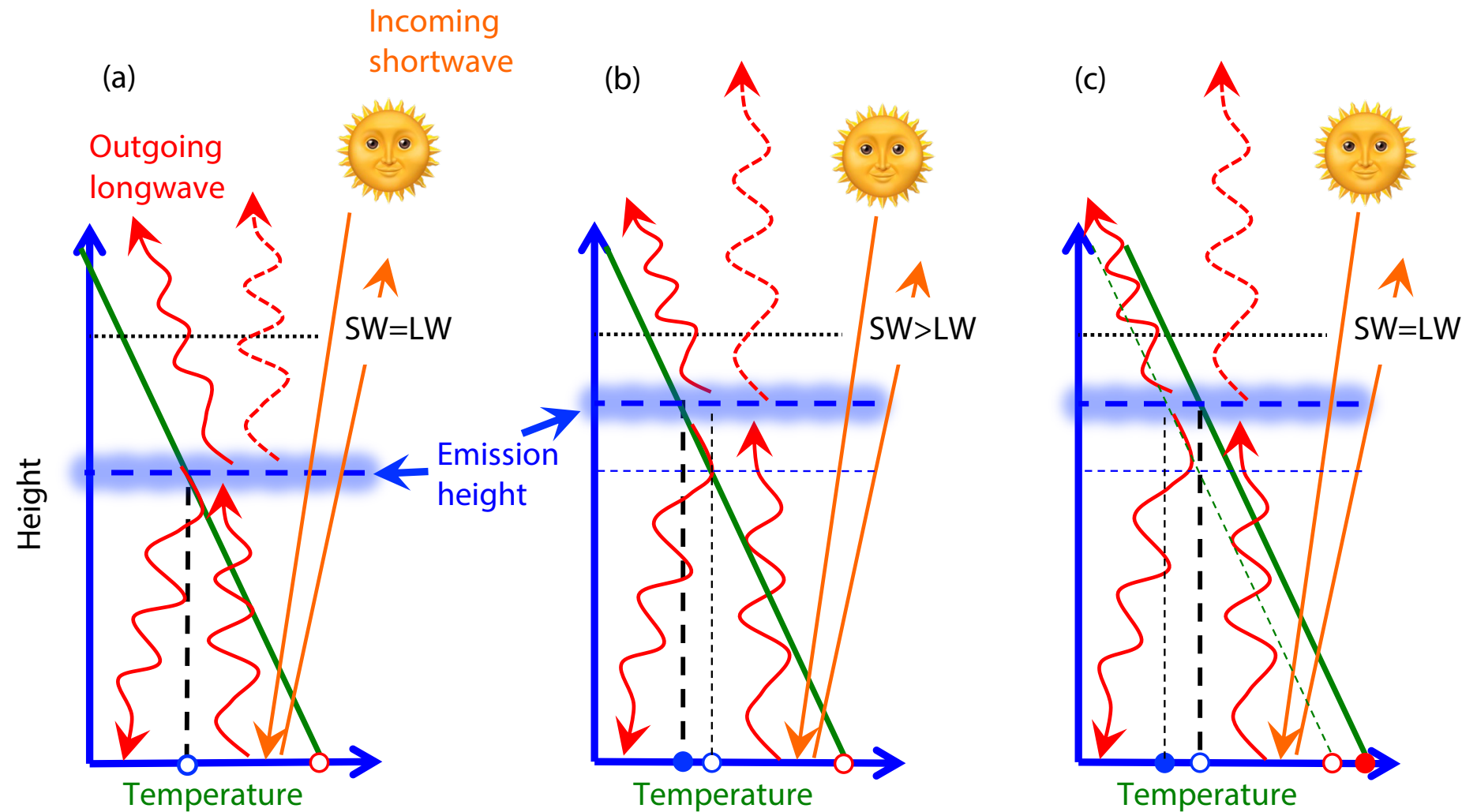
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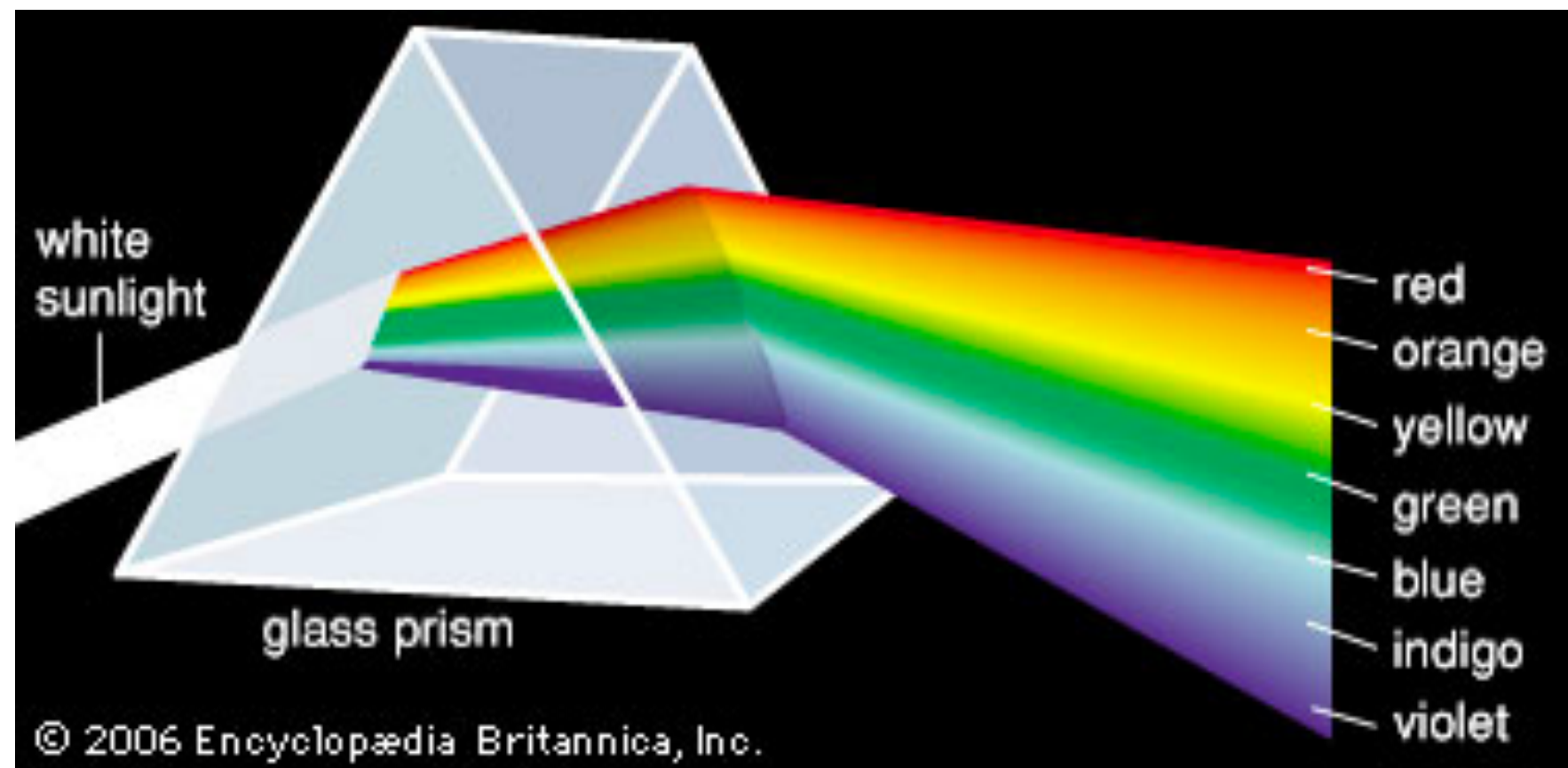


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Workshop #2 c:
emission height: leave for HW

How greenhouse gasses work:
wavelength-dependent radiation & molecular energy levels

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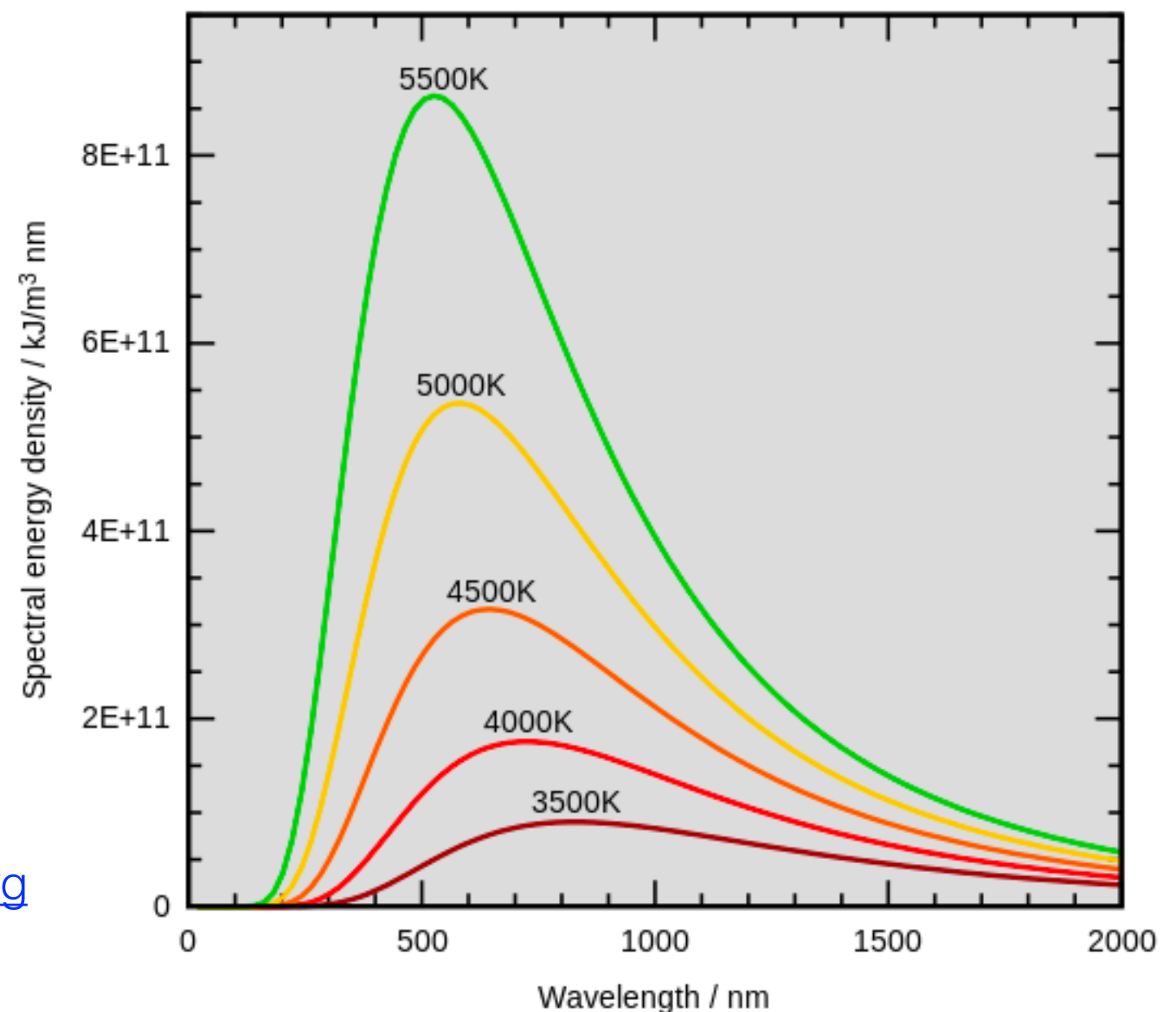
Black body radiation

Planck's law of black-body radiation:
$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

$B(\lambda, T)d\lambda$ is the energy per area/ time/ angle emitted between wavelengths λ & $\lambda+d\lambda$; T =temperature; h =Planck's const; c =speed of light; k =Boltzmann's const.

Total emitted radiation per area/ time: σT^4 Stefan–Boltzmann constant:

$\sigma = 5.670367 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$



https://en.wikipedia.org/wiki/File:Wiens_law.svg

Shortwave vs longwave radiation reminder

Earth's surface and the sun both emit blackbody radiation according to Planck's function — they radiate over the full spectrum

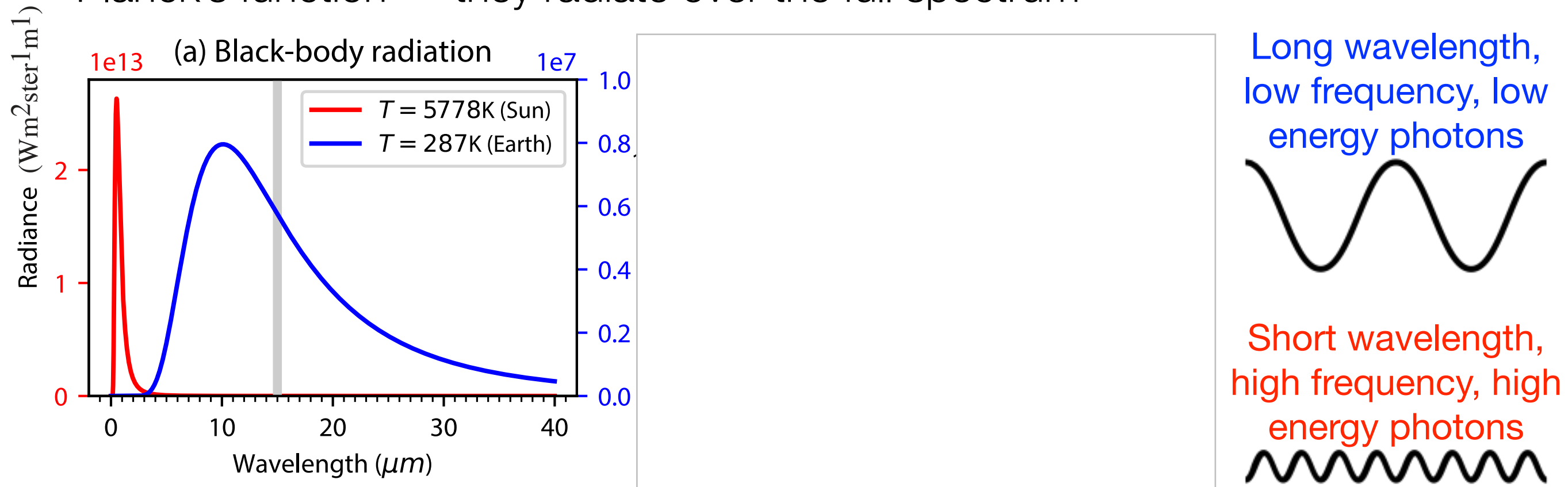


Figure 2.5: Blackbody radiation.

(a) Planck's black-body spectral radiance for the emission temperatures of the Earth (blue) and Sun (red), as a function of wavelength.

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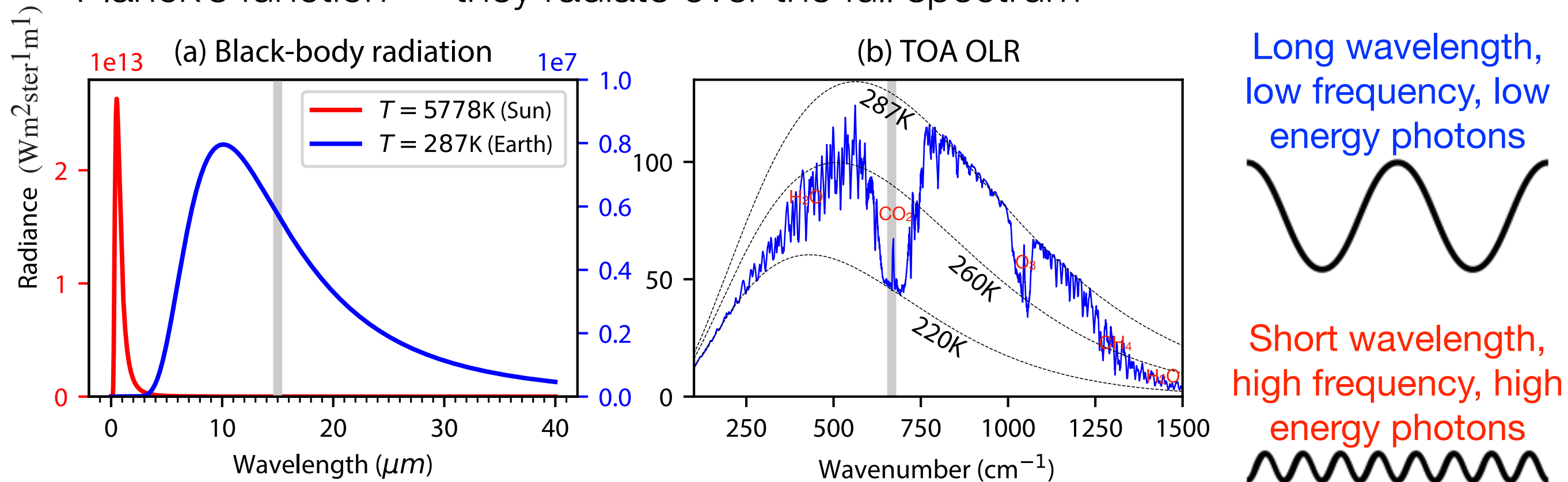
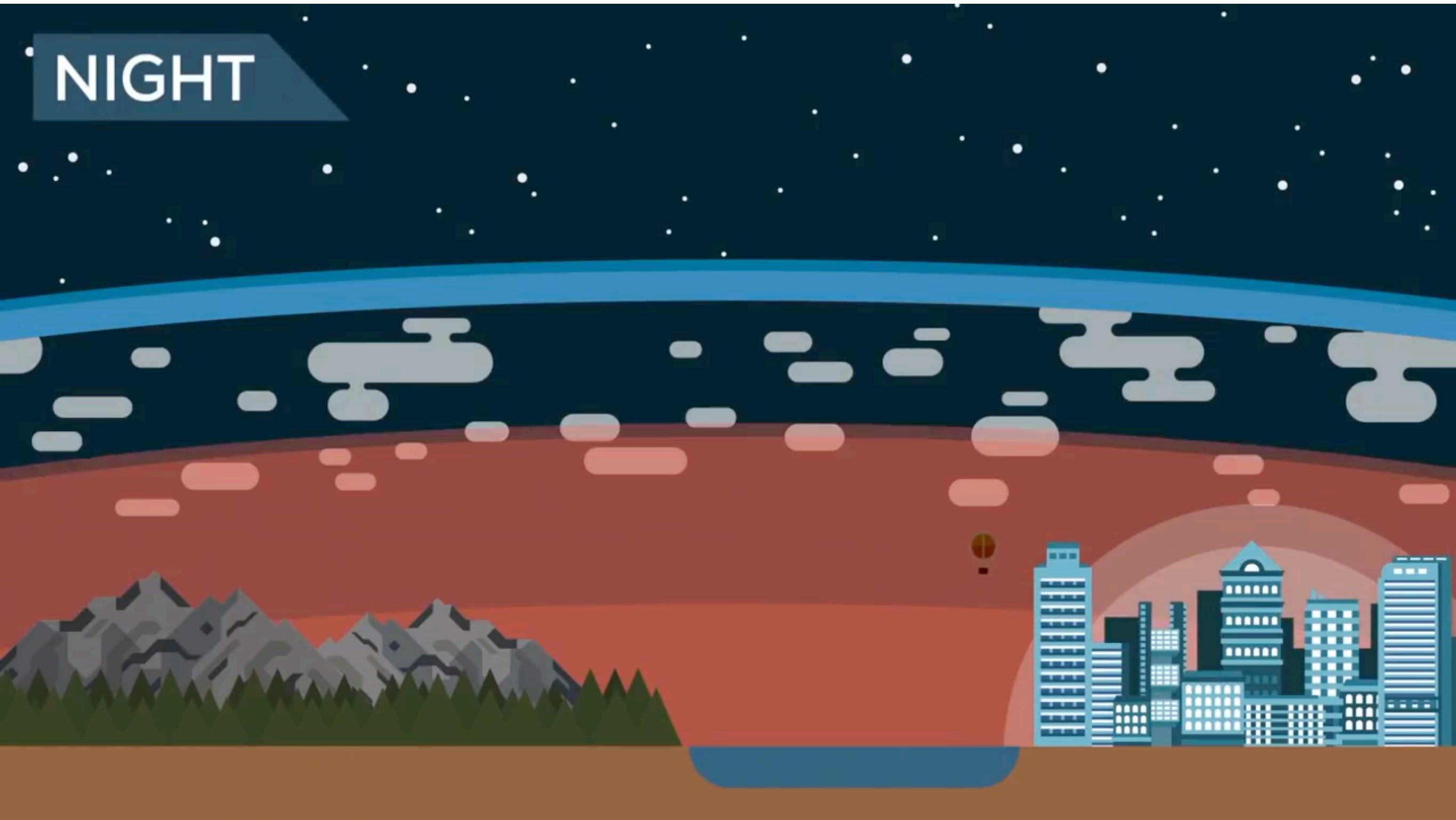


Figure 2.5: Blackbody radiation.

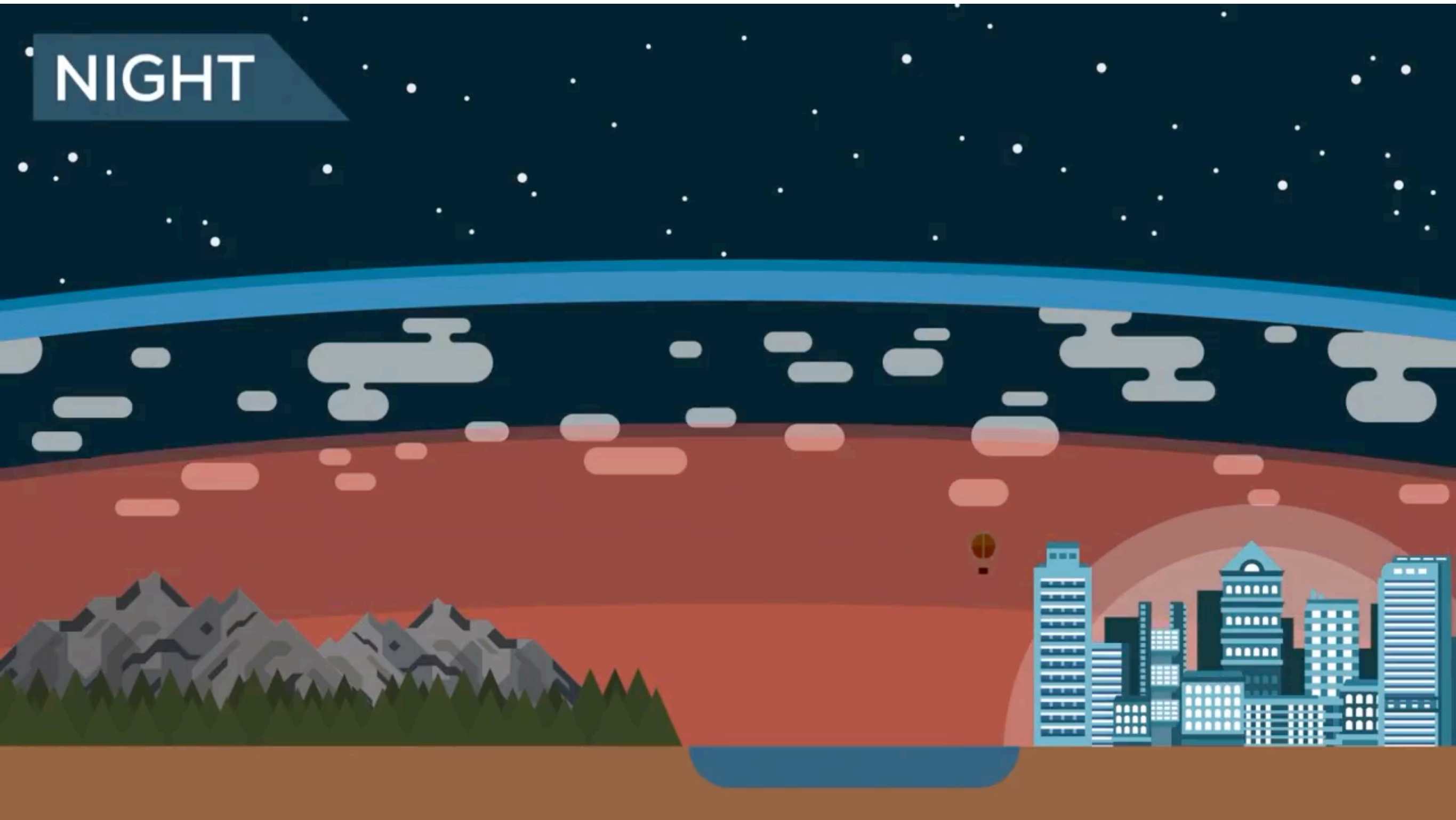
(a) Planck's black-body spectral radiance for the emission temperatures of the Earth (blue) and Sun (red), as a function of wavelength.

(b) blue: estimated outgoing longwave radiation at the top of the atmosphere as a function of wavenumber; dashed lines: black-body radiation curves at different temperatures. The deviations from the 287 K black body radiation curve indicate absorption bands due to CO_2 , CH_4 , H_2O , and O_3 ; the central CO_2 absorption line is shown on both panels as a vertical gray bar.

How do greenhouse gases work?

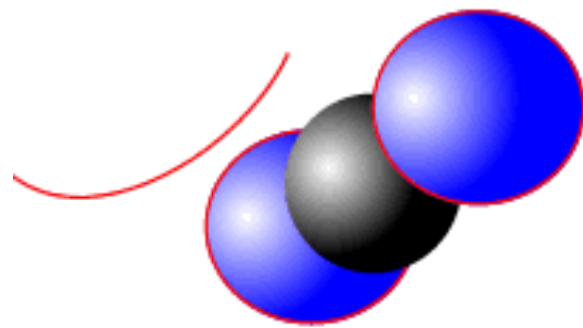


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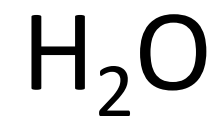
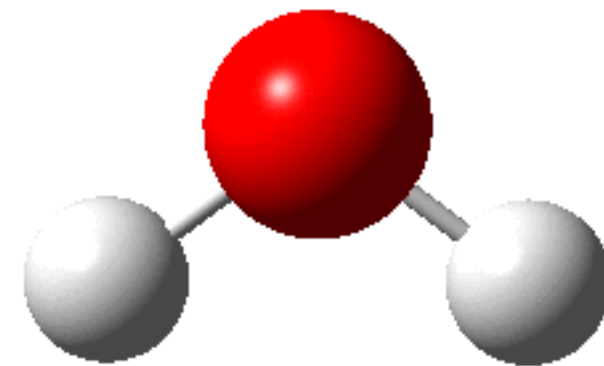


<https://youtu.be/sTvqlijqvTg>

Vibration energy levels of CO₂ & H₂O



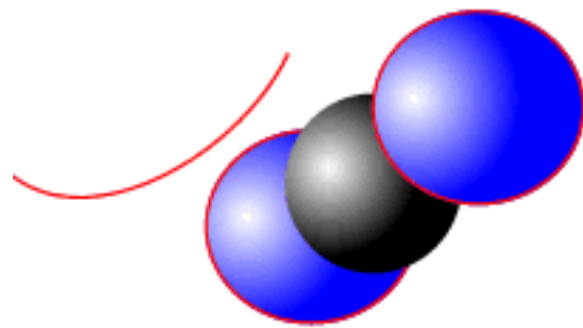
<http://www.dynamicscience.com.au/tester/solutions1/chemistry/greenhouse/co2andtheghe.htm>



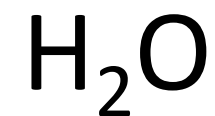
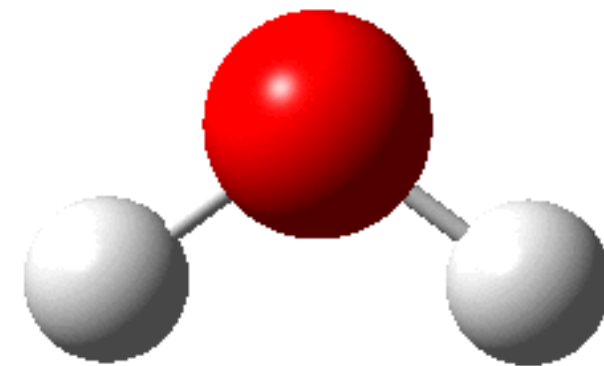
<http://davidobru.blogspot.com/2017/01/some-animations.html>

The vibration energy levels determine the frequency of absorption

Vibration energy levels of CO₂ & H₂O



<http://www.dynamicscience.com.au/tester/solutions1/chemistry/greenhouse/co2andtheghe.htm>

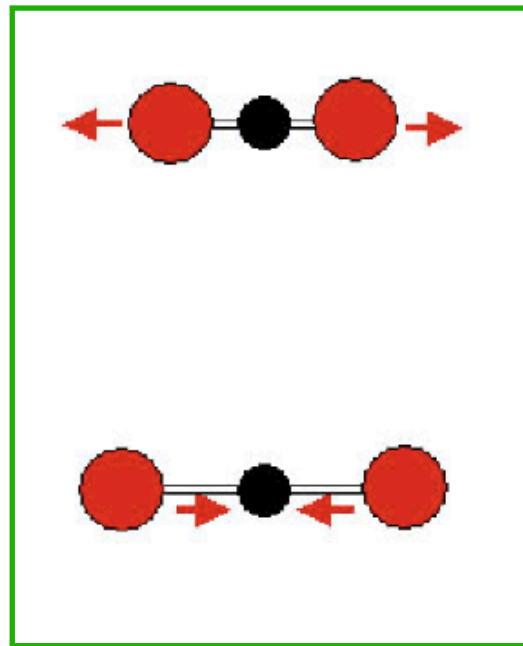


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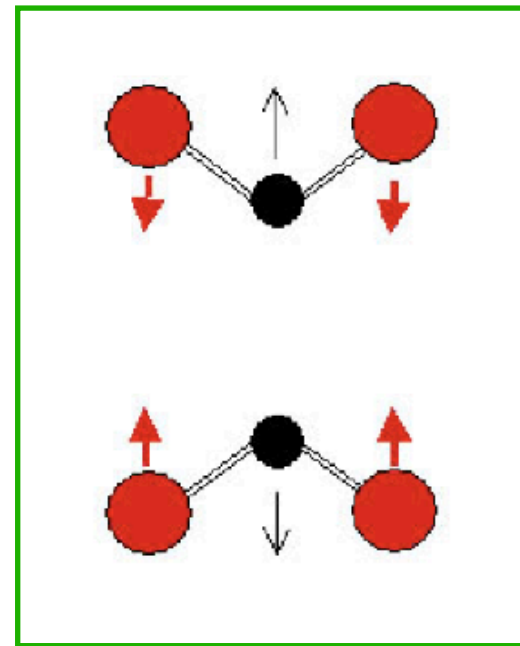
The vibration energy levels determine the frequency of absorption

Greenhouse gasses vibration modes

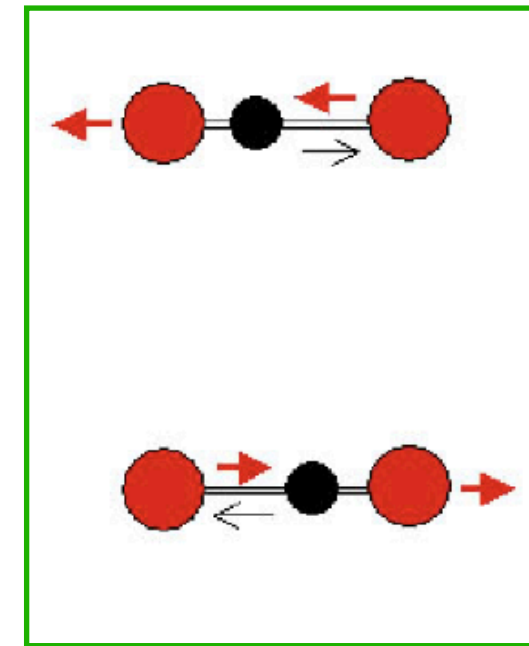
- Discrete wavelengths excite molecular transitions in greenhouse gas molecules
- This results in absorption and re-emission of radiation of that wavelength
- The “transition wavelengths” for various greenhouse gases have been measured in lab experiments



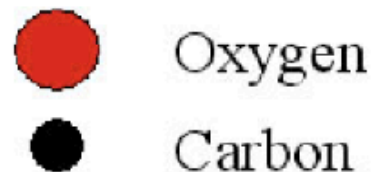
Symmetric Stretch
1366 cm^{-1}



Bending Mode
667 cm^{-1}

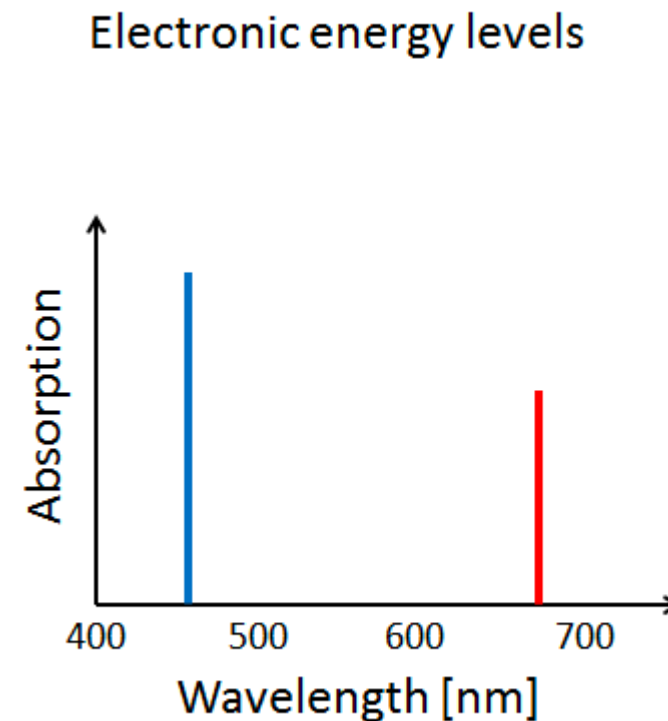
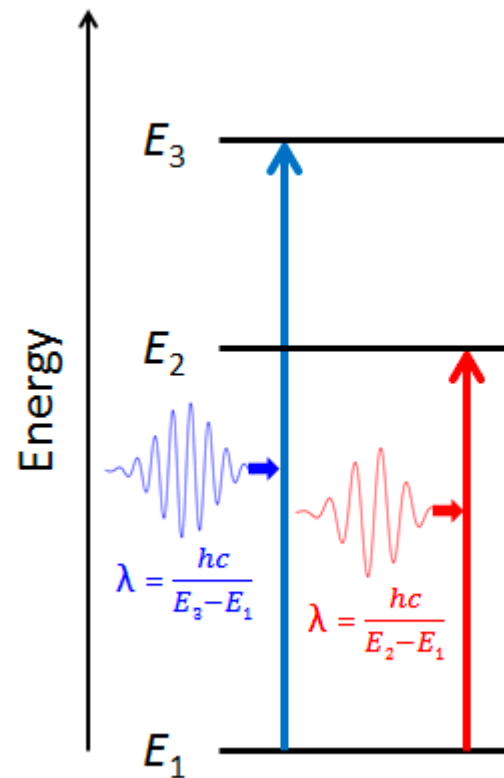


Asymmetric Stretch
2349 cm^{-1}



<http://butane.chem.uiuc.edu/pshapley/GenChem1/L15/2.html>

Molecular vibration/rotation energy levels & photon absorption

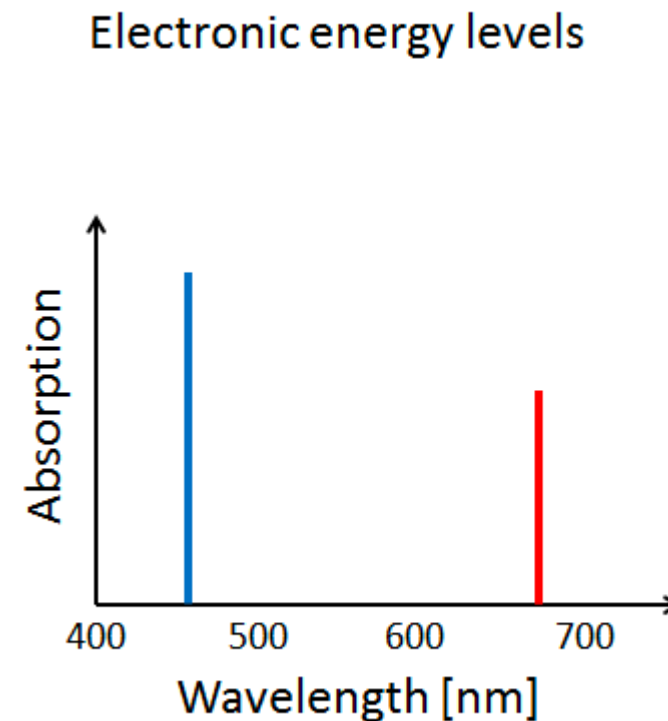
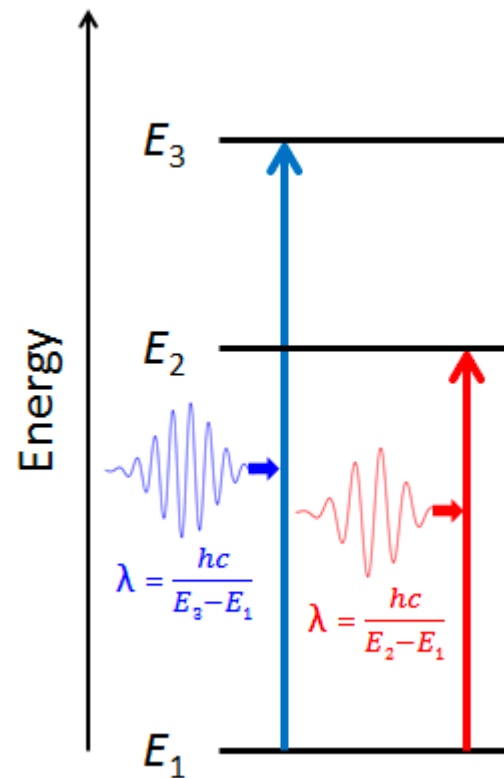


Photon energy: $E = h\nu = hc/\lambda$ where h is Planck's constant, c is the speed of light, ν the frequency, and λ the wavelength. Note: $\nu = c/\lambda = 1/T$, where T is the wave period.

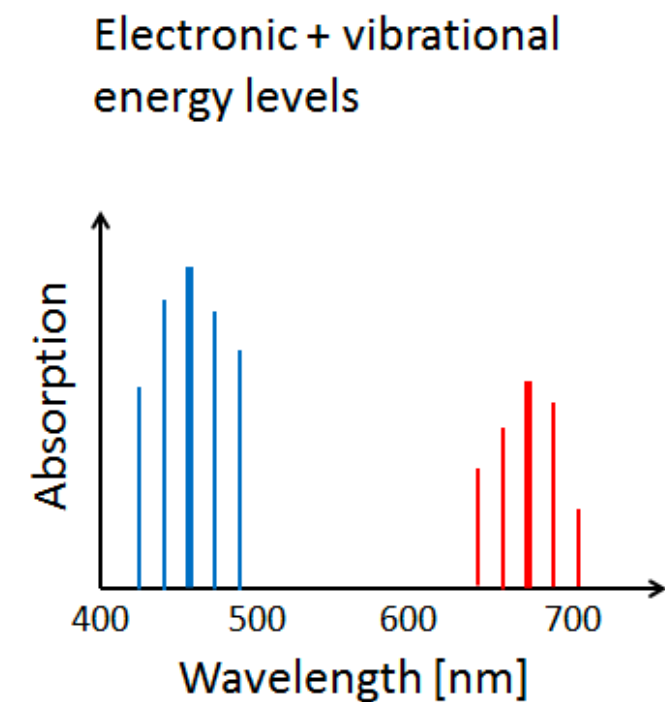
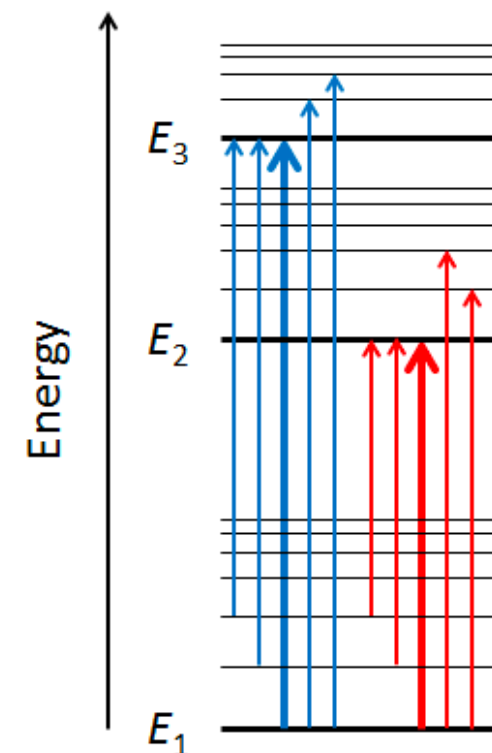
From "ocean optics" web book,

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Molecular vibration/rotation energy levels & photon absorption



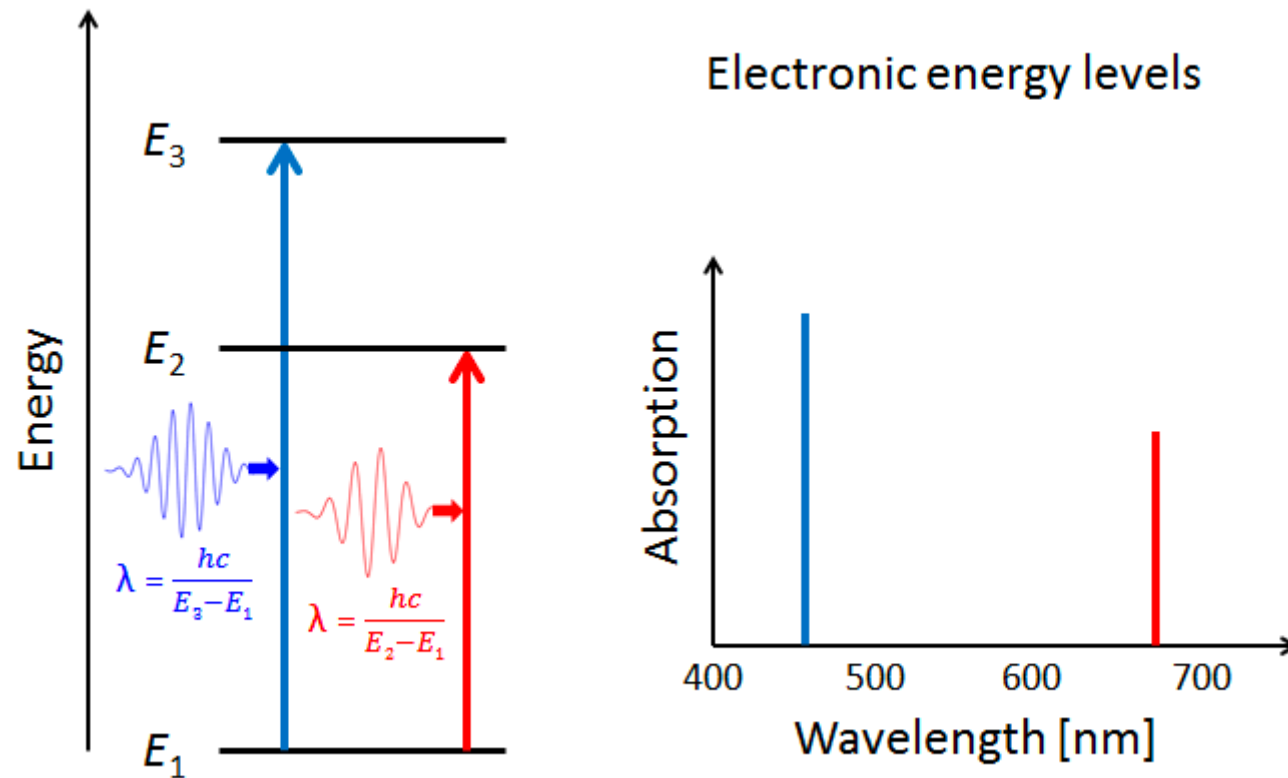
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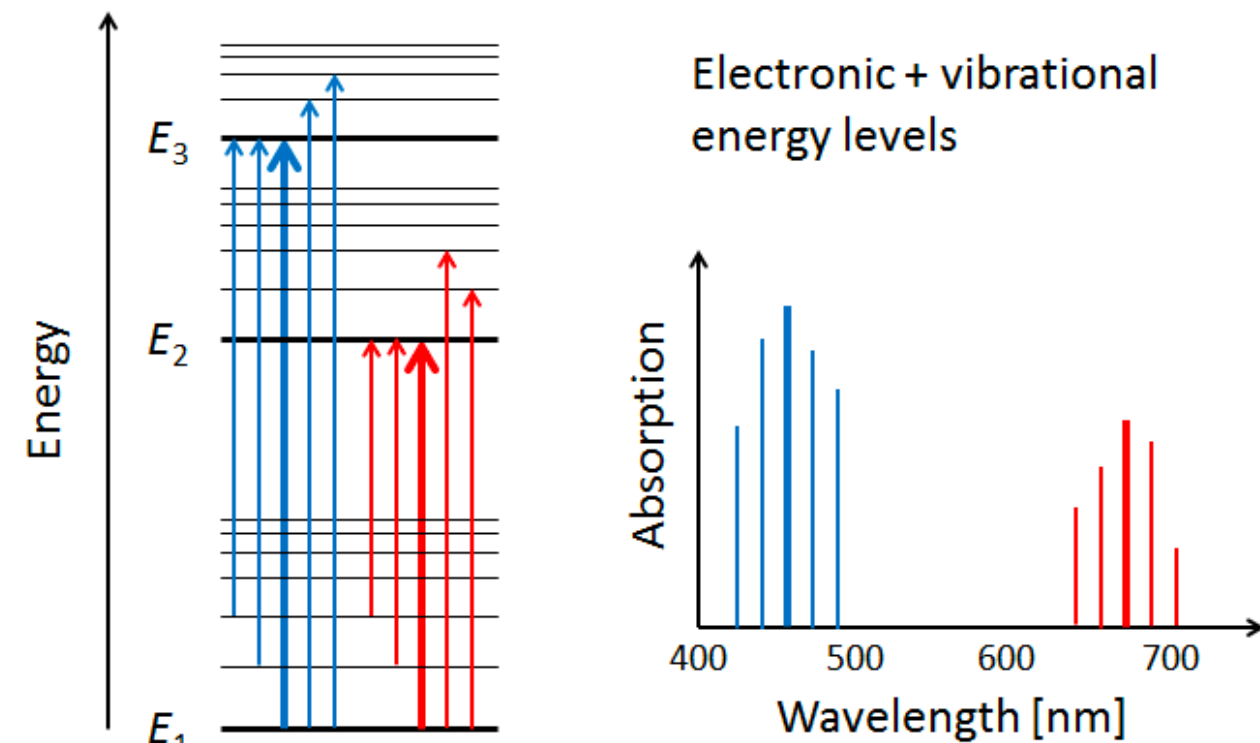
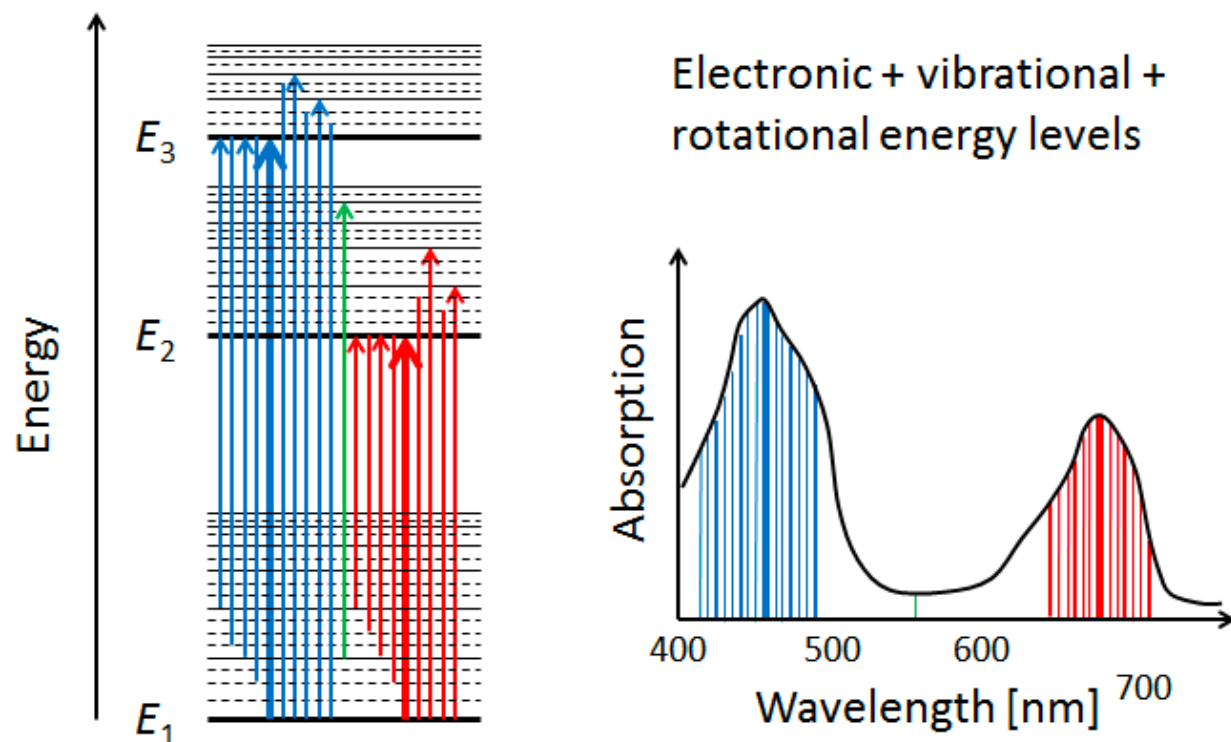
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Molecular vibration/rotation energy levels & photon absorption



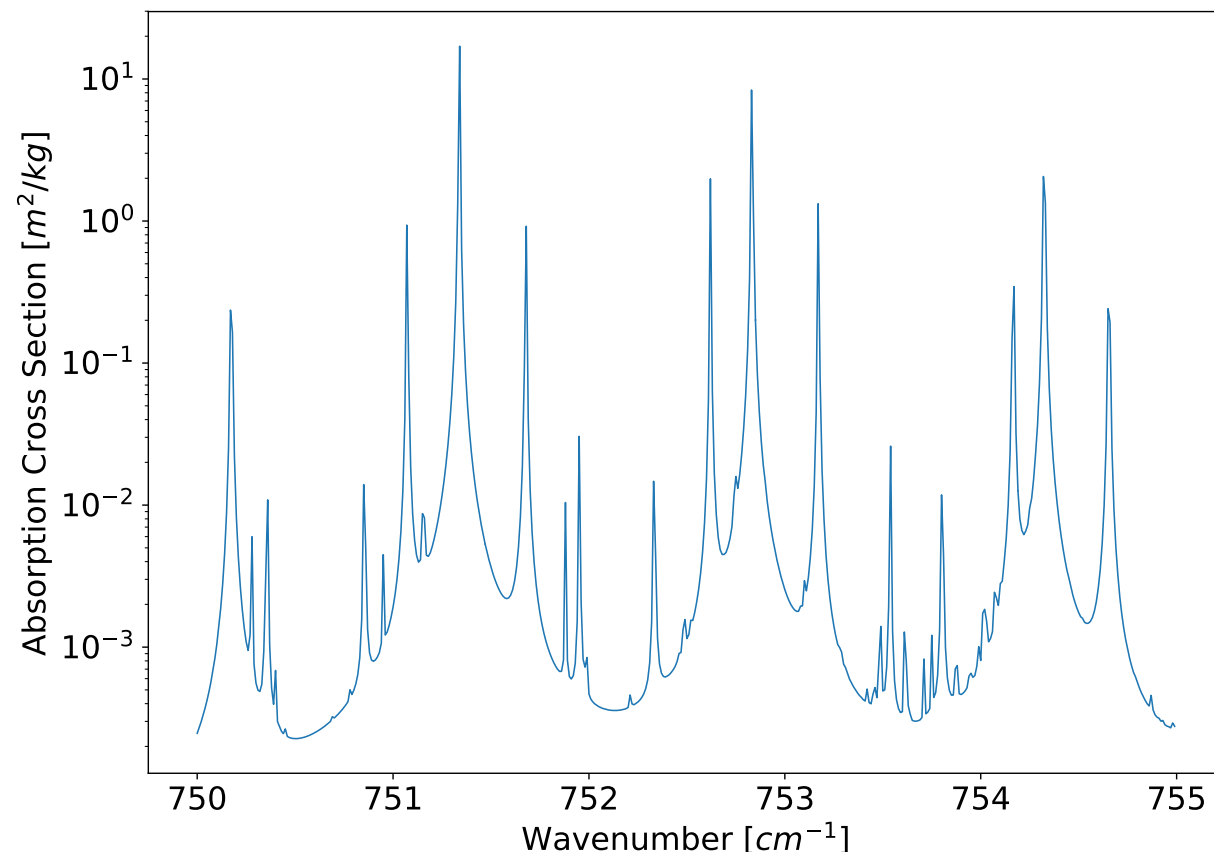
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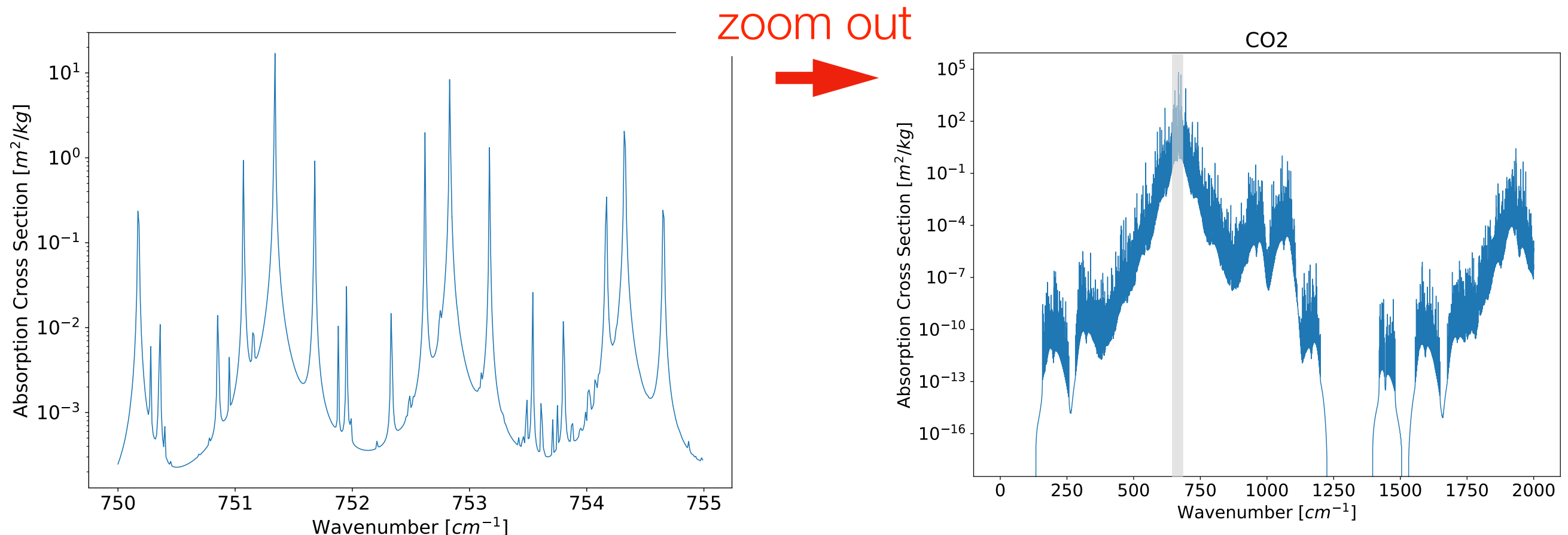
Absorption cross-section

- Spike in absorption “efficiency” i.e. cross-section at wavenumber that excites a molecular transition: spikes called absorption lines
- Absorption “lines” look like Gaussian/Lorentzian $(\alpha_L/\pi)/((\nu - \nu_0)^2 + \alpha_L^2)$ profiles due to “line broadening”
 - Pressure broadening due to molecular collisions
 - Doppler broadening due to gaussian distribution of particle velocities
- All individual lines are measured, and then broadened in radiative transfer code to give “absorption spectrum” for a greenhouse gas



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Notes section 2.2.3:
Broadening

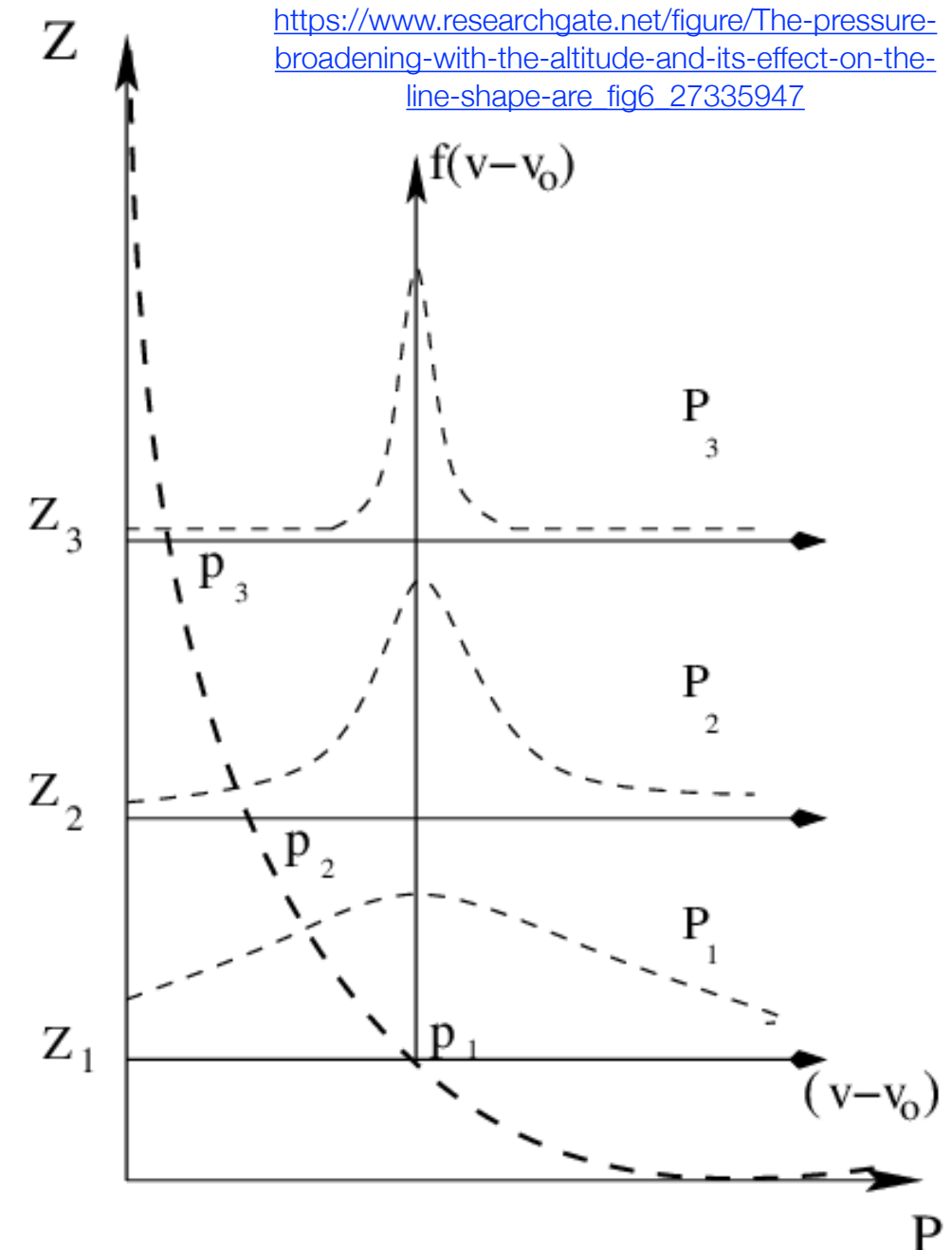
(use next three slides)

Pressure/Collisional Broadening

Low pressure,
fewer collisions

High pressure,
more collisions

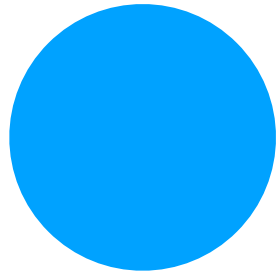
Absorption efficiency



The collisions of gas molecules cause the widening of absorption lines:

If the arriving photon has slightly more energy than needed for energy level transition, the excess energy can be transferred to the colliding molecule, allowing to absorb photons that are not exactly at the right frequency/ energy

Pressure/Collisional Broadening

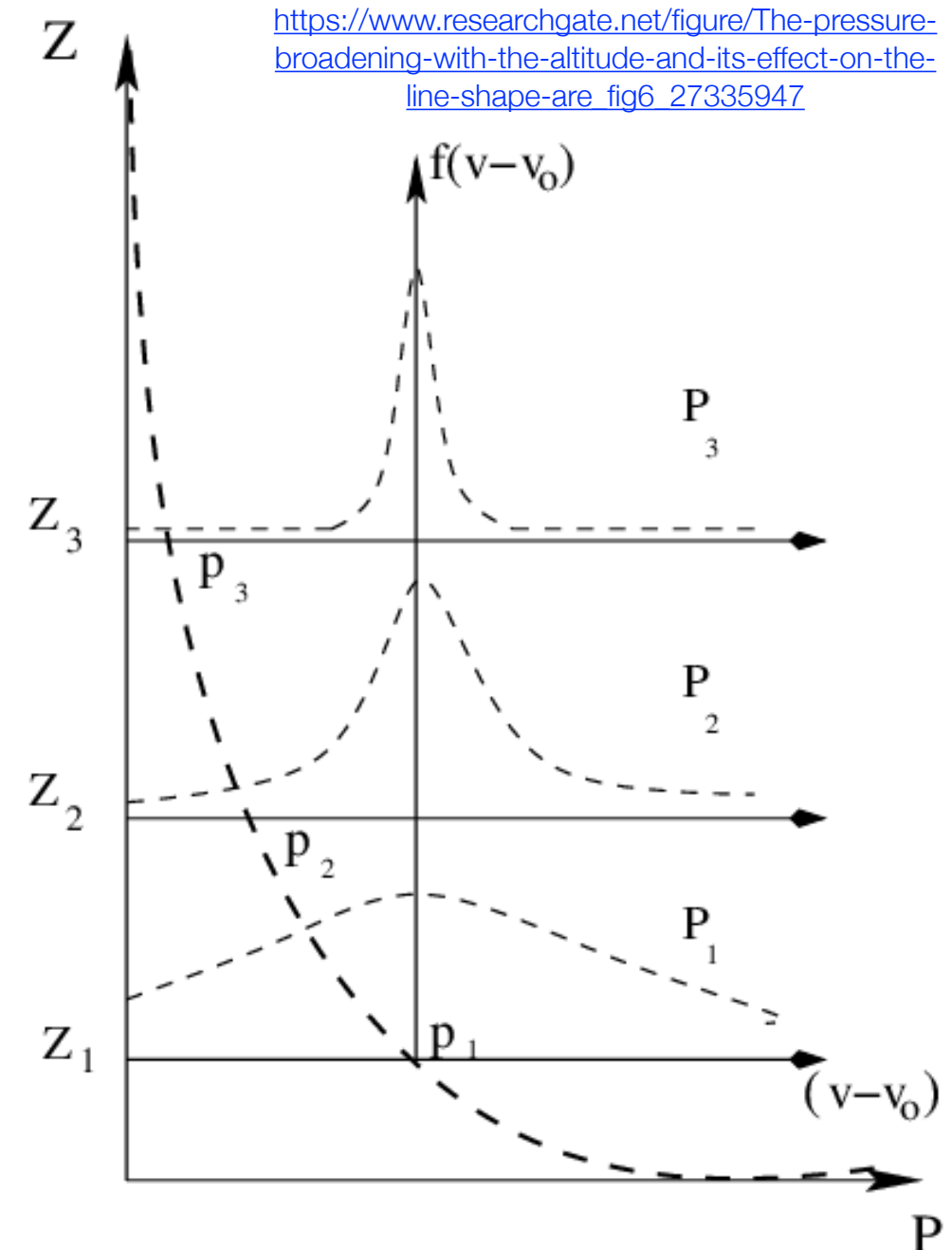


Low pressure,
fewer collisions

↑

High pressure,
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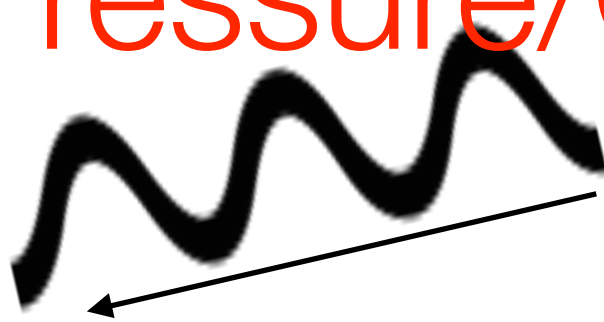
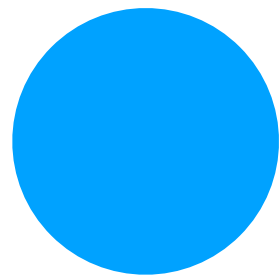
Absorption efficiency



The collisions of gas molecules cause the widening of absorption lines:

If the arriving photon has slightly more energy than needed for energy level transition, the excess energy can be transferred to the colliding molecule, allowing to absorb photons that are not exactly at the right frequency/ energy

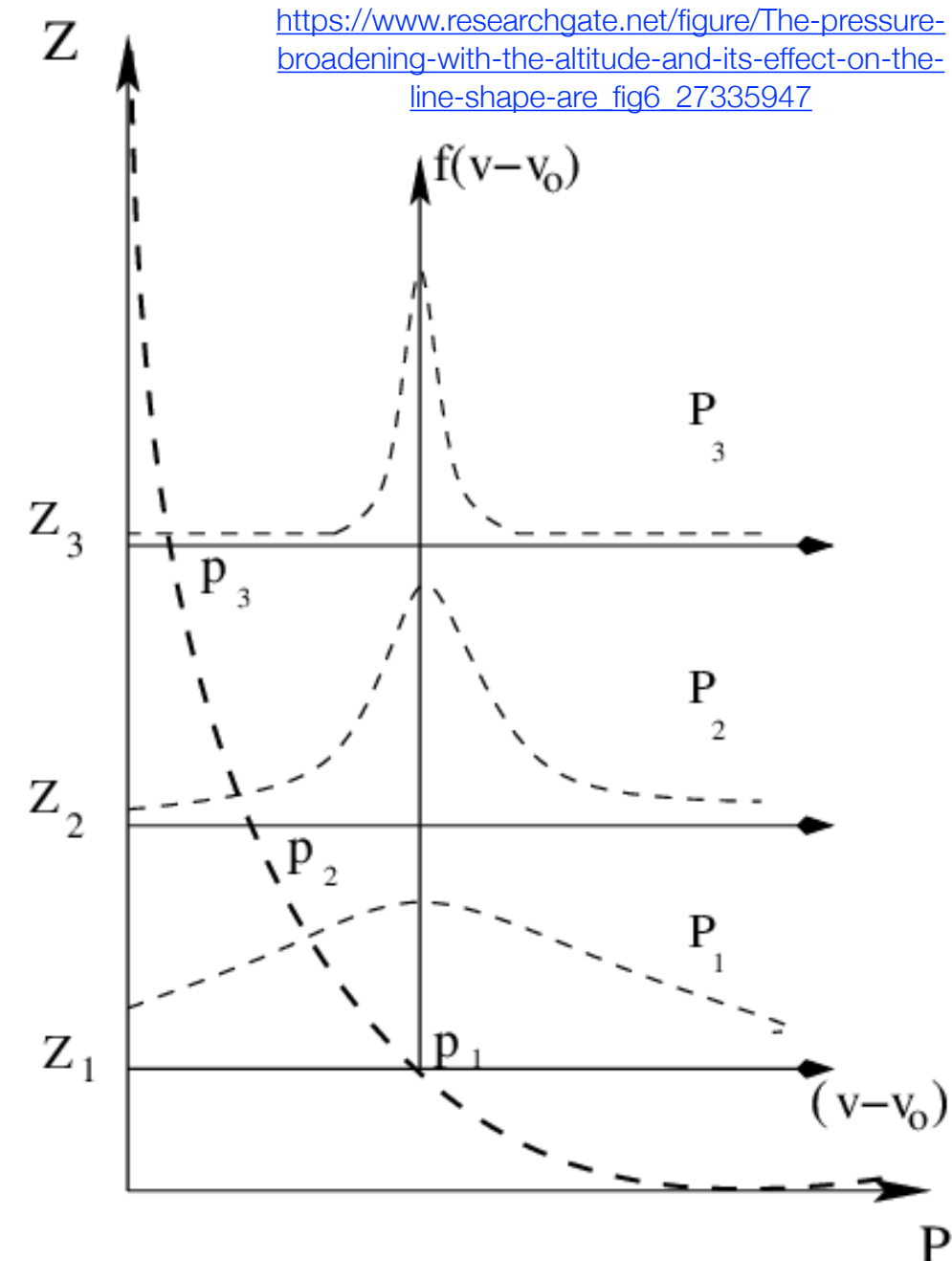
Pressure/Collisional Broadening



Low pressure,
fewer collisions

High pressure,
more collisions

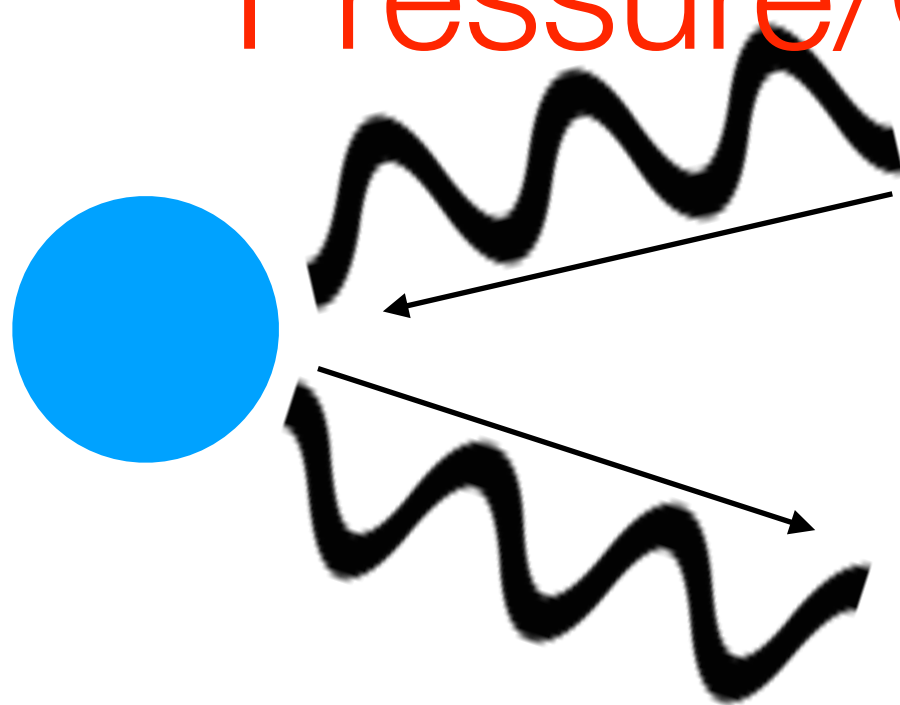
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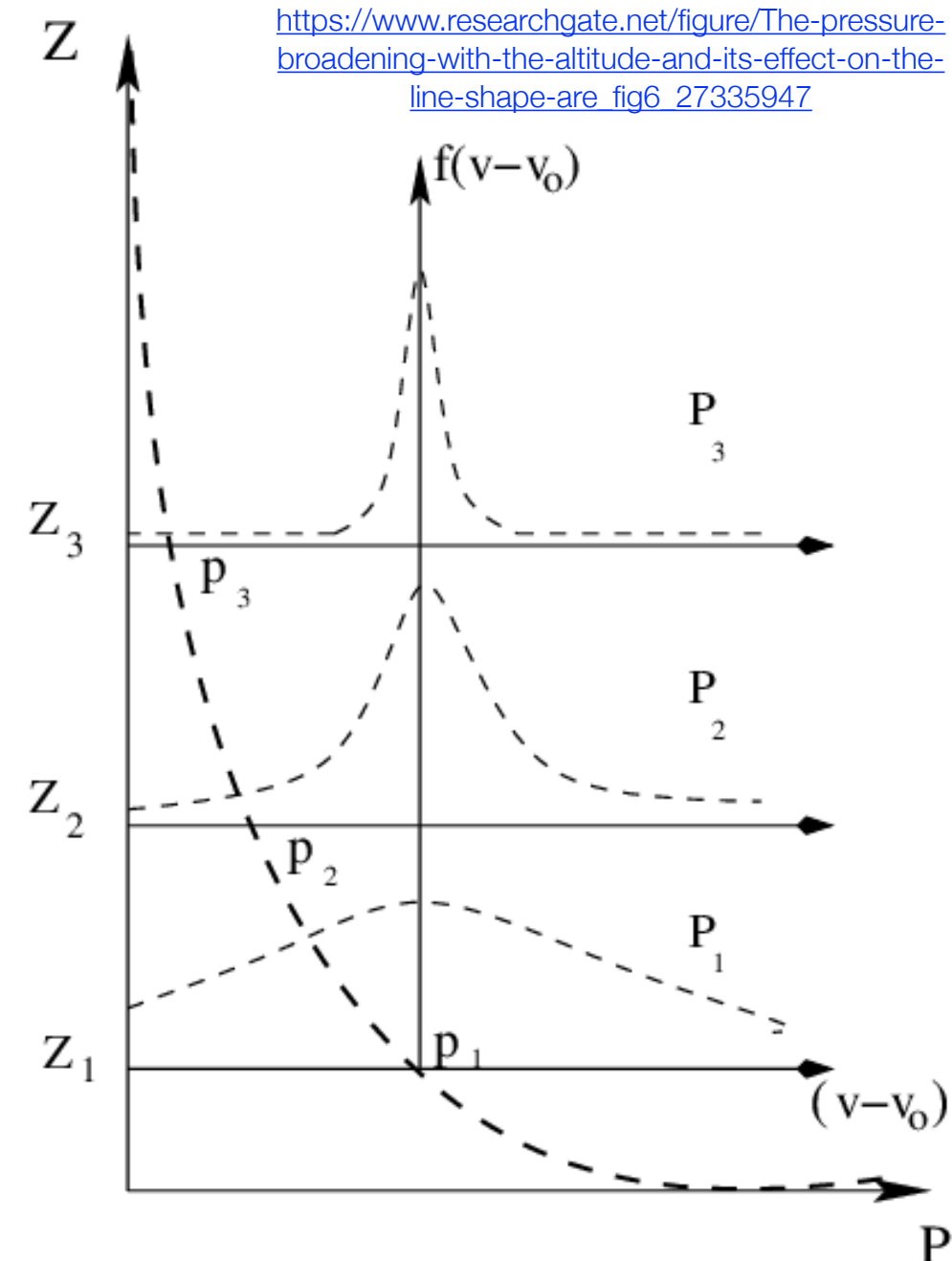
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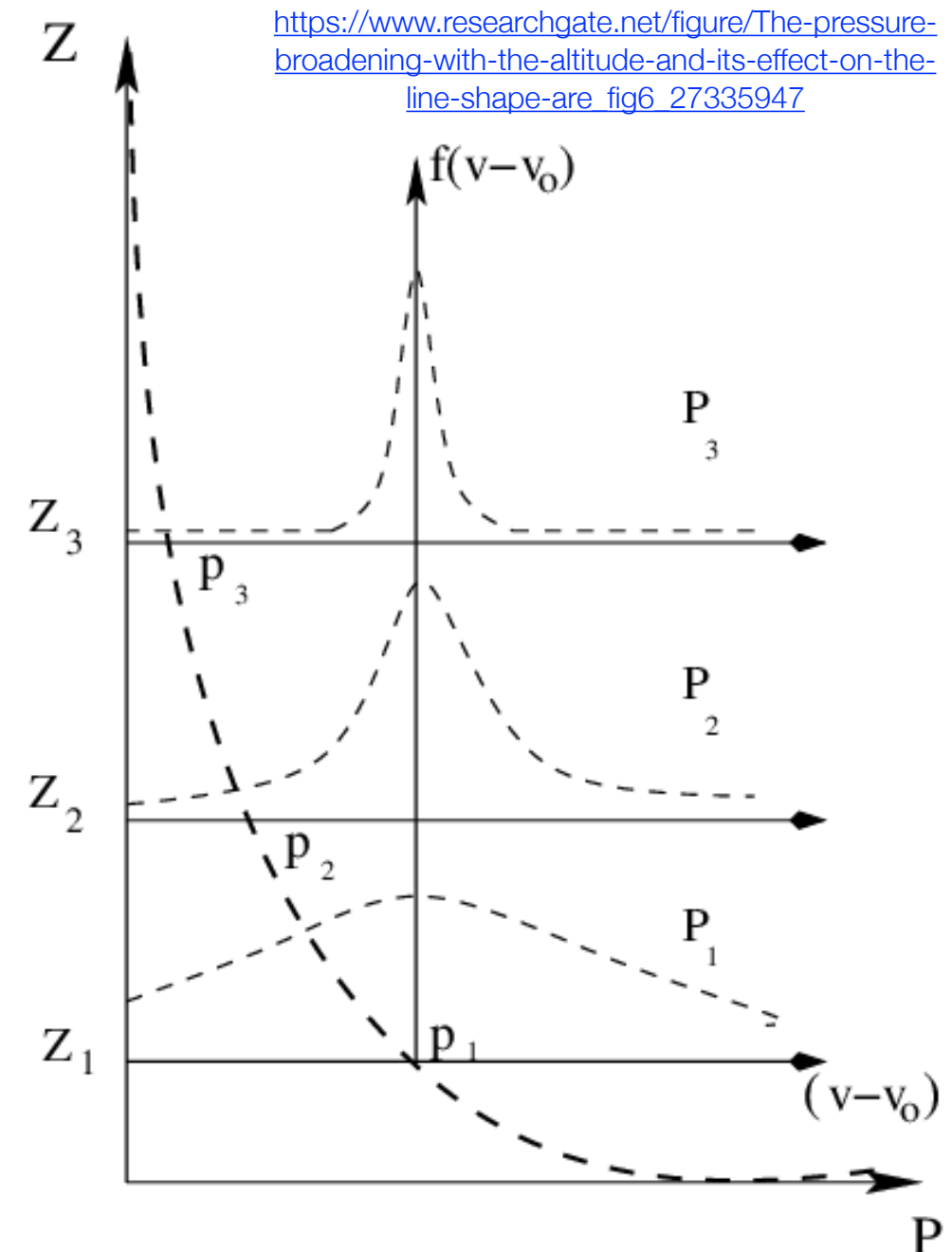
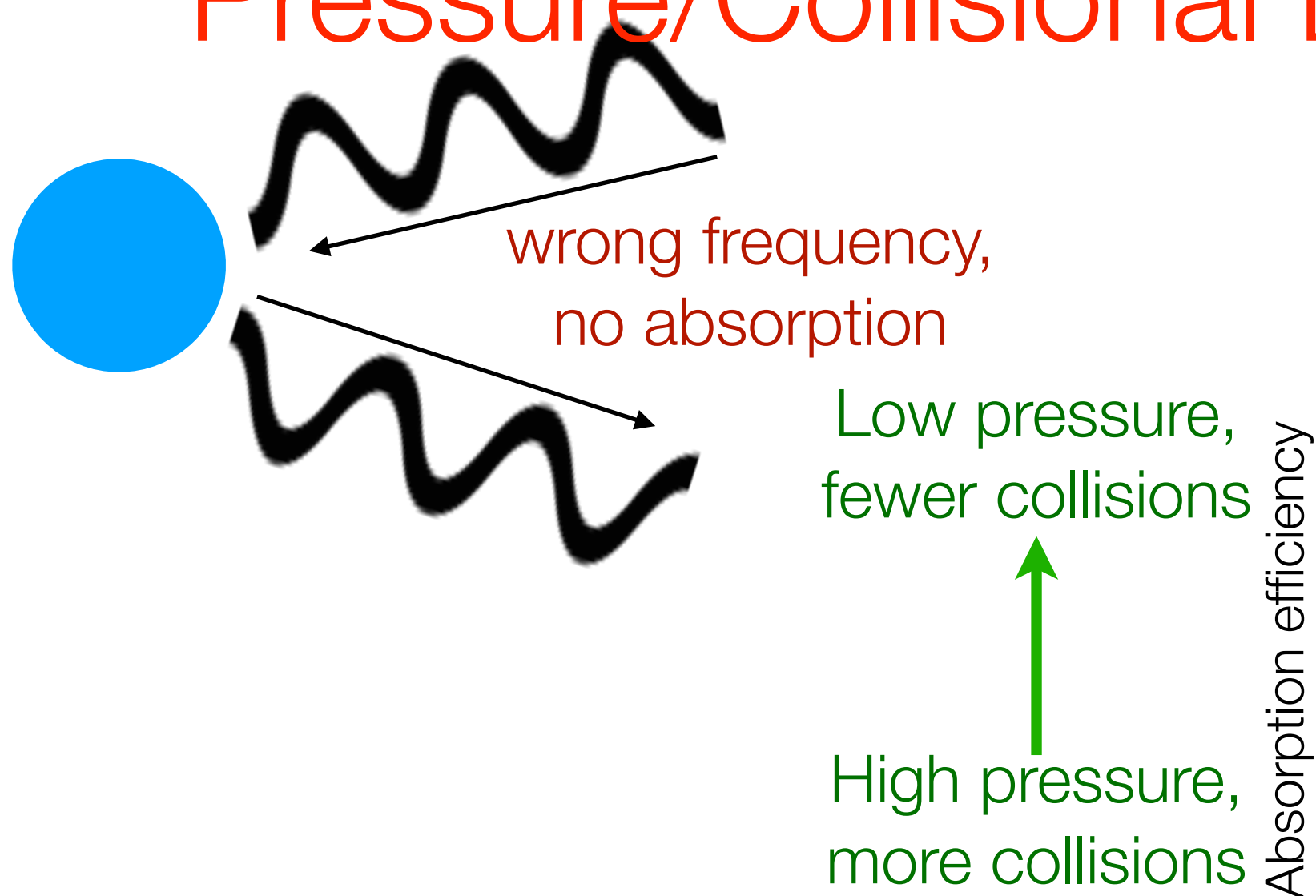
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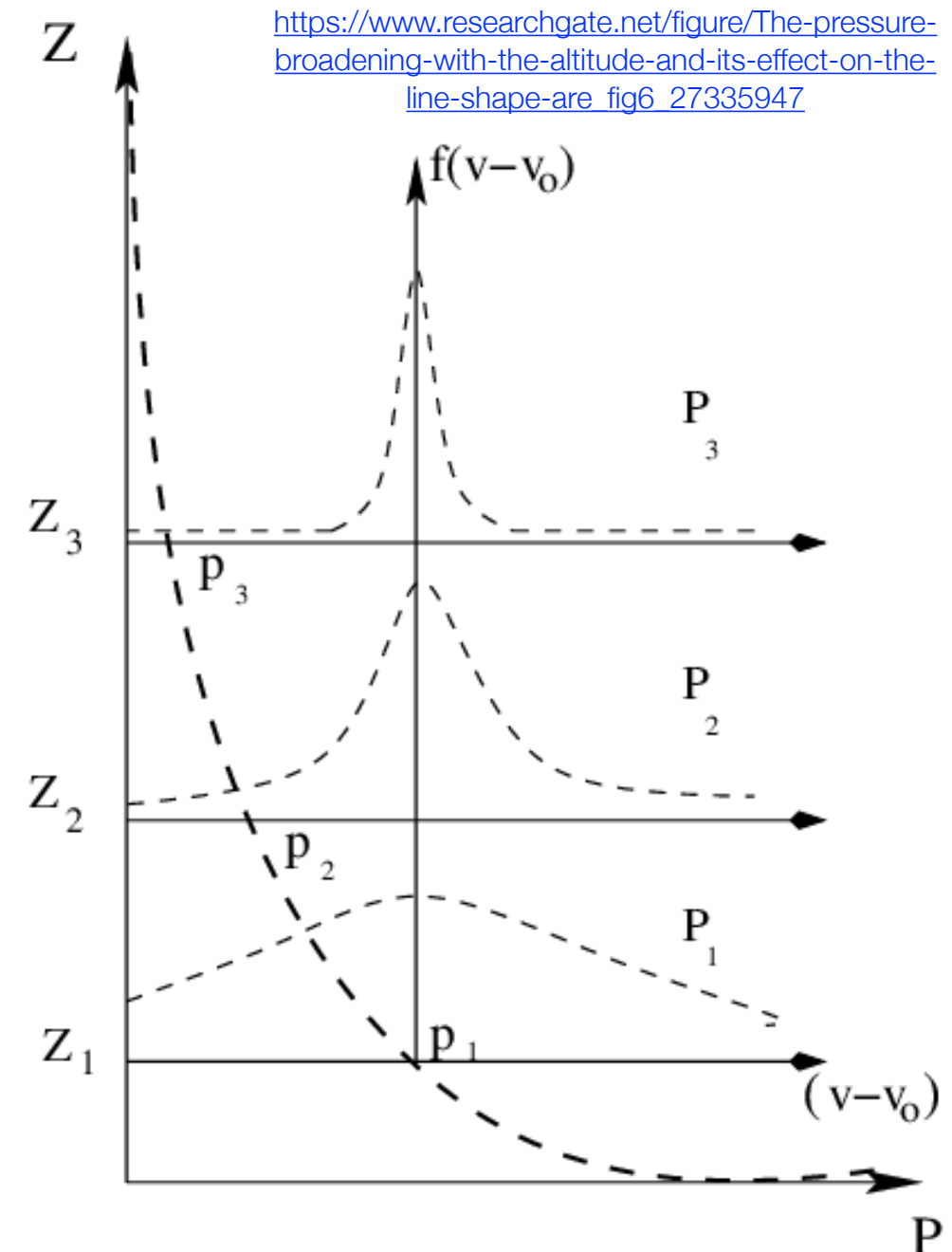
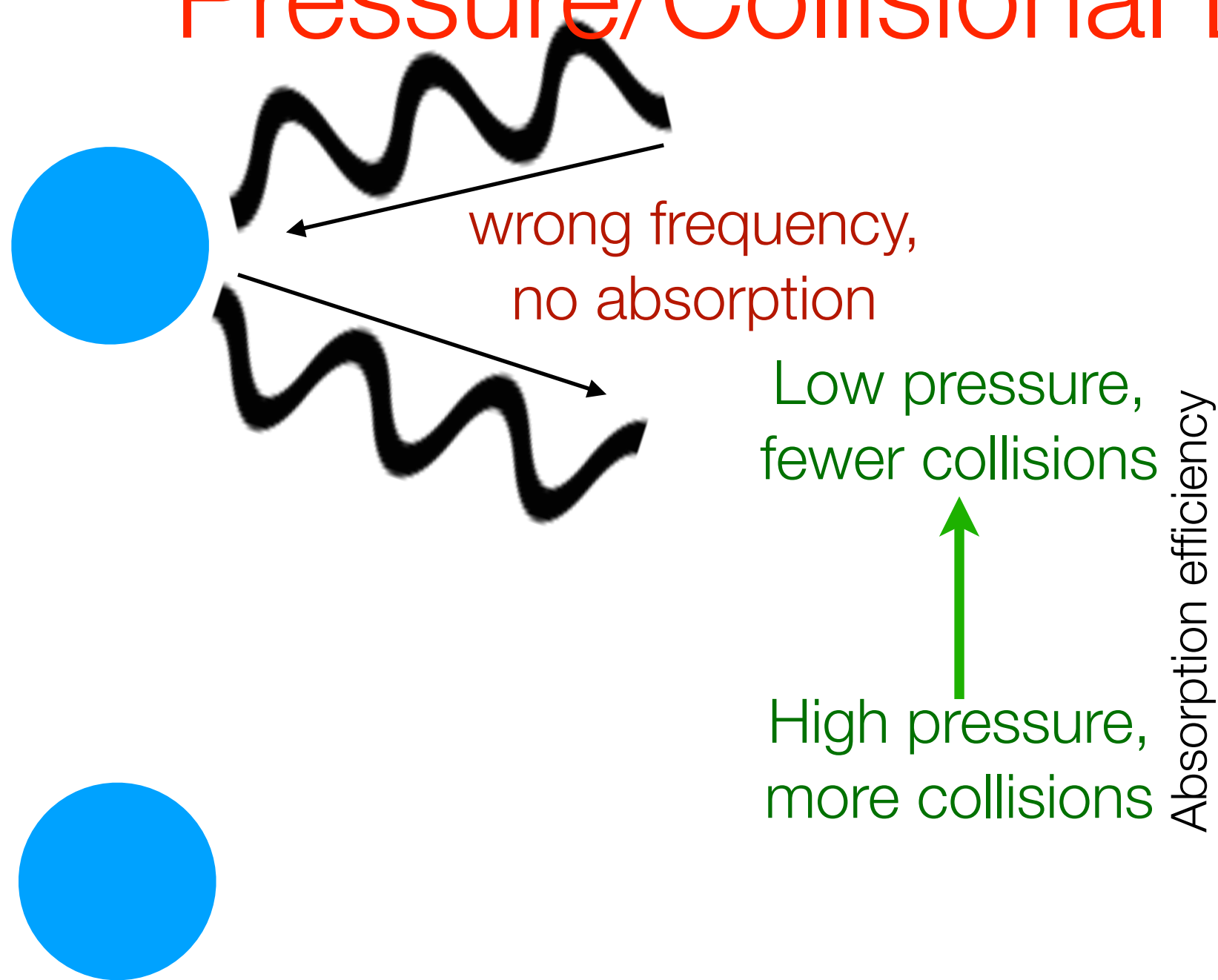
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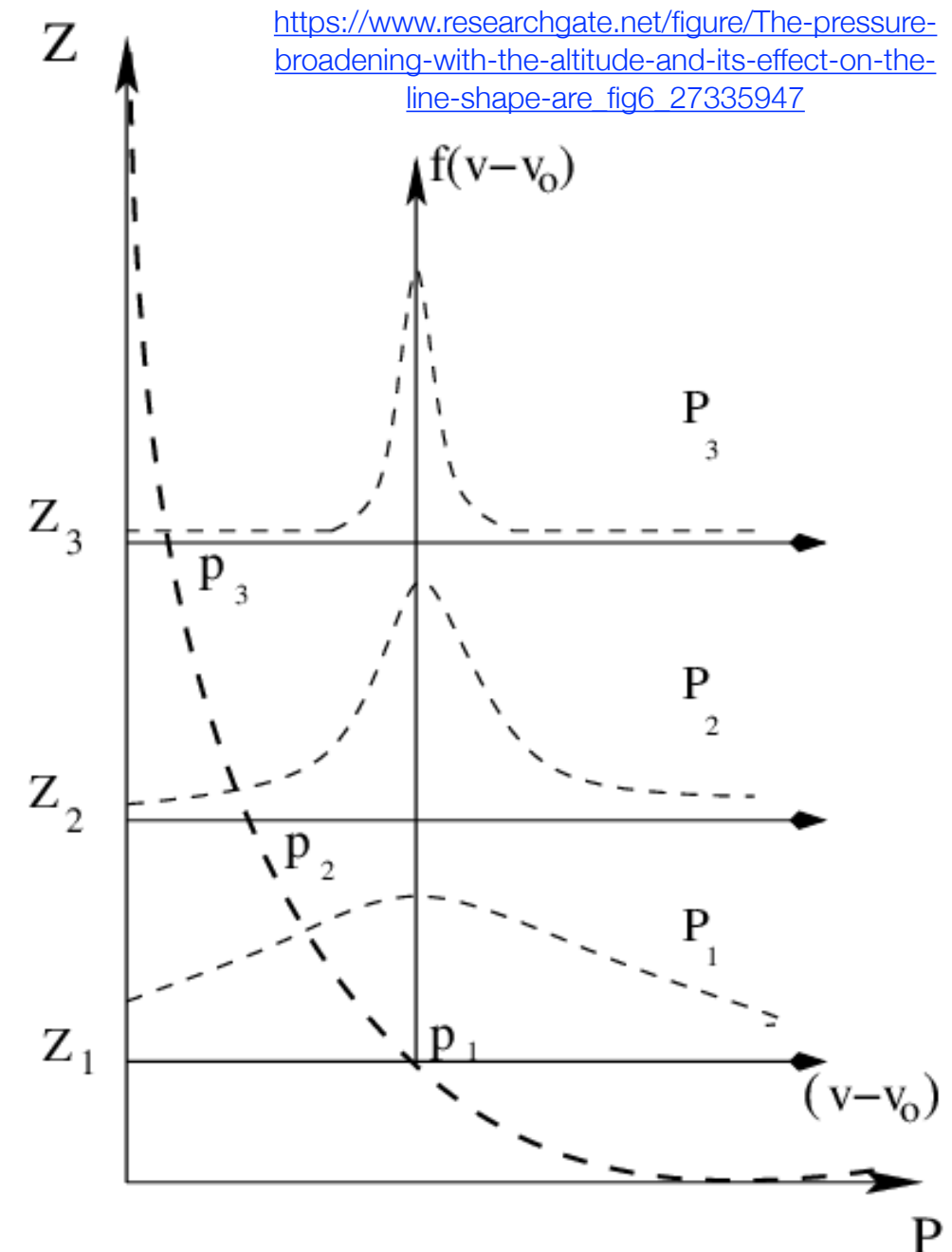
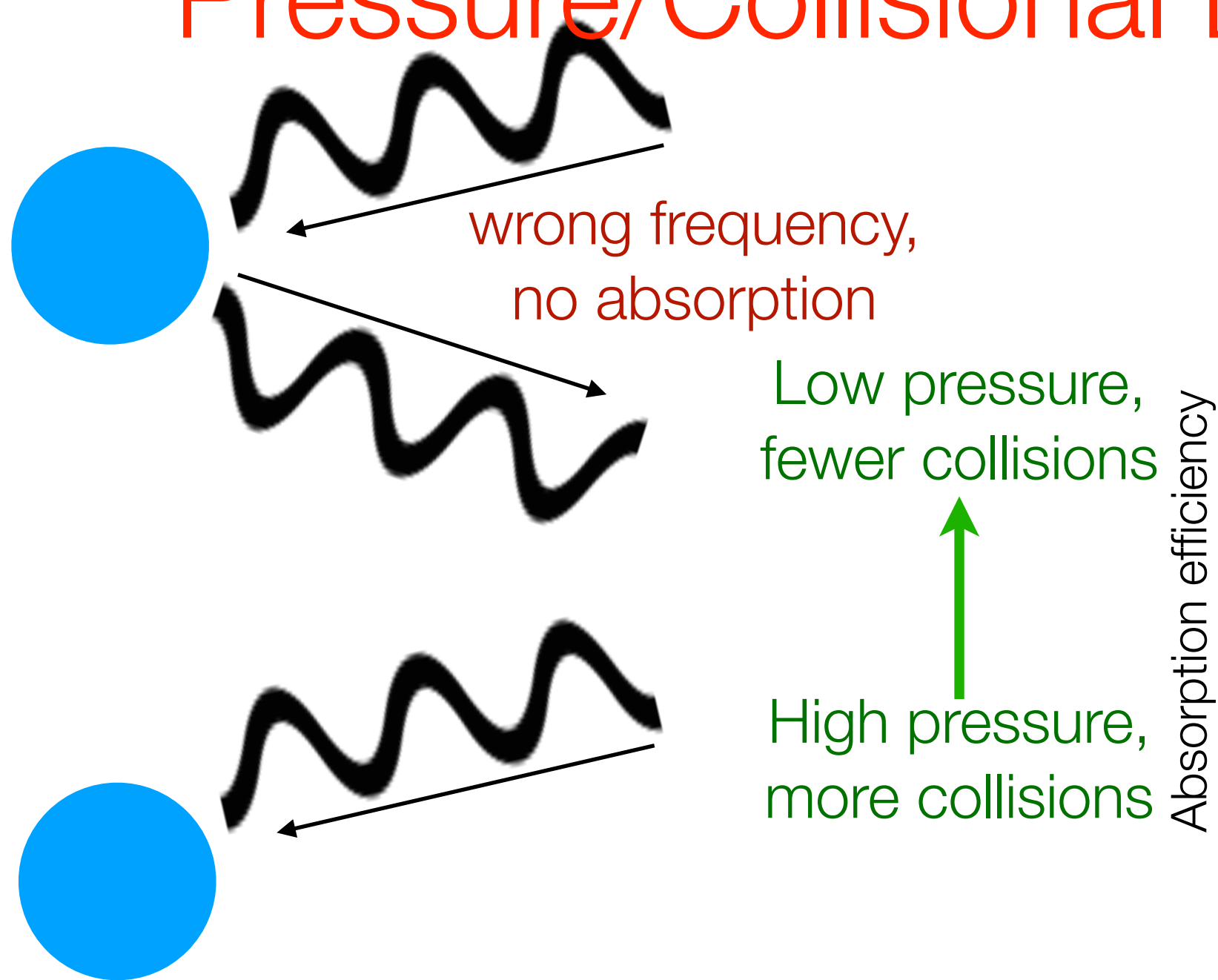
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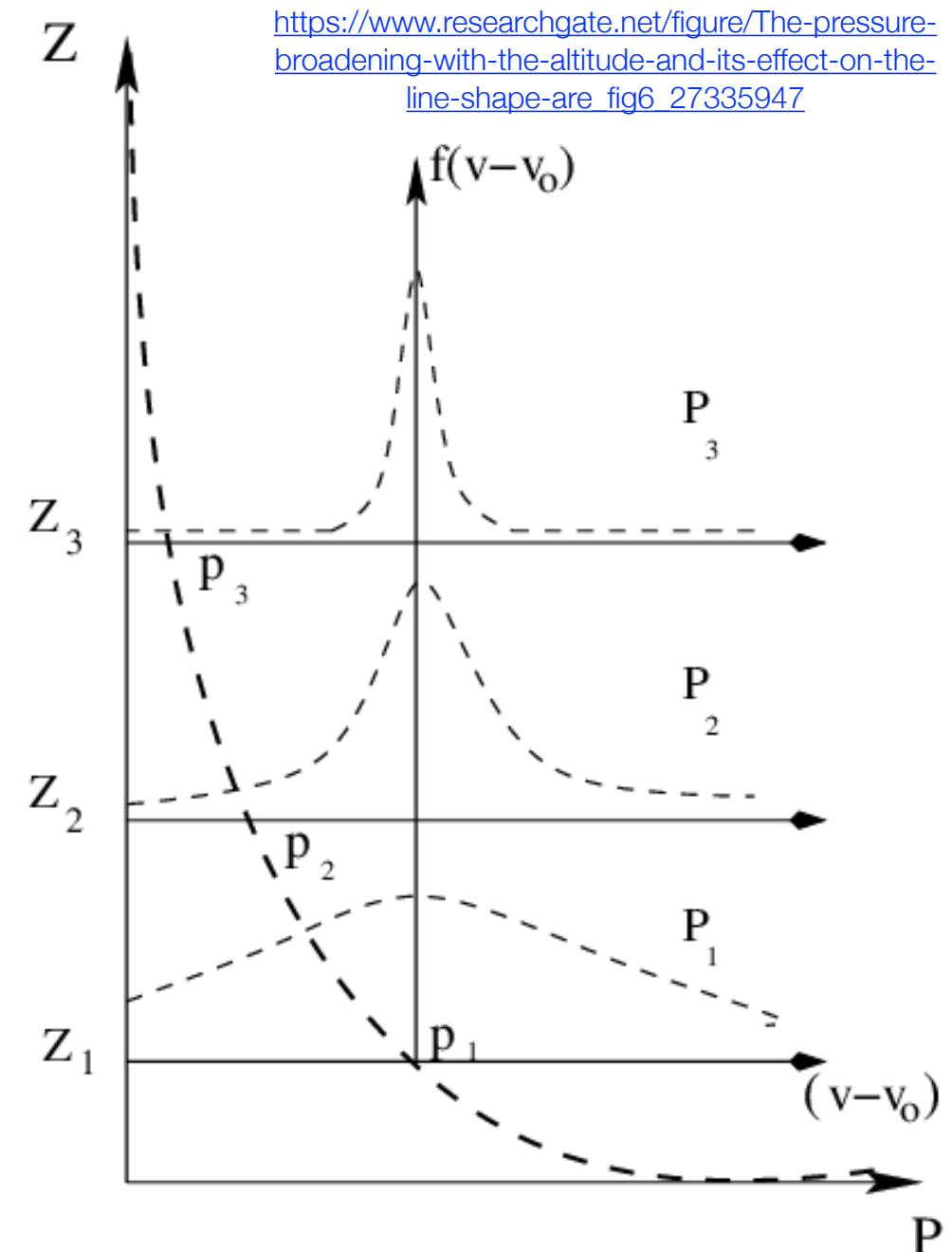
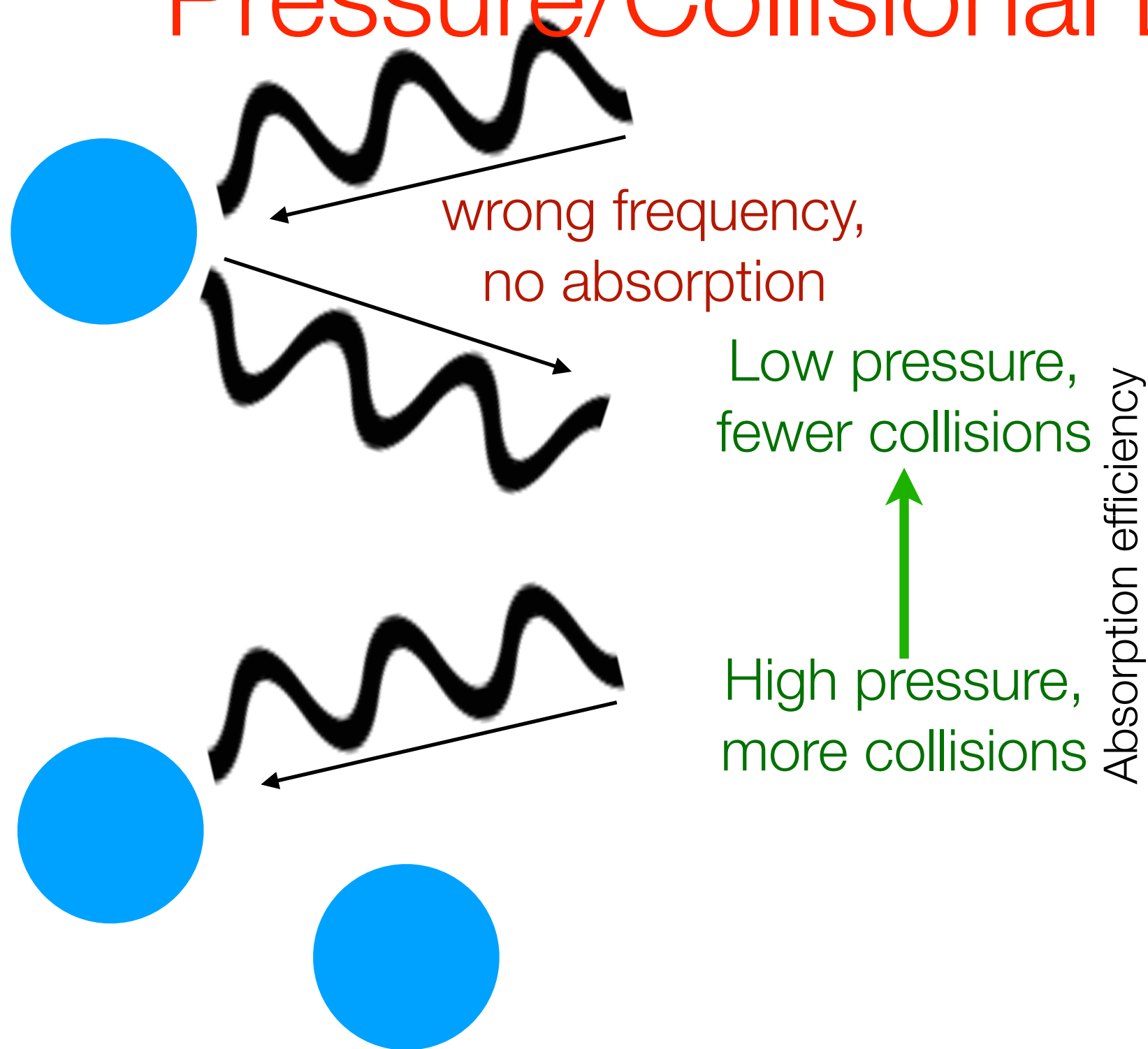
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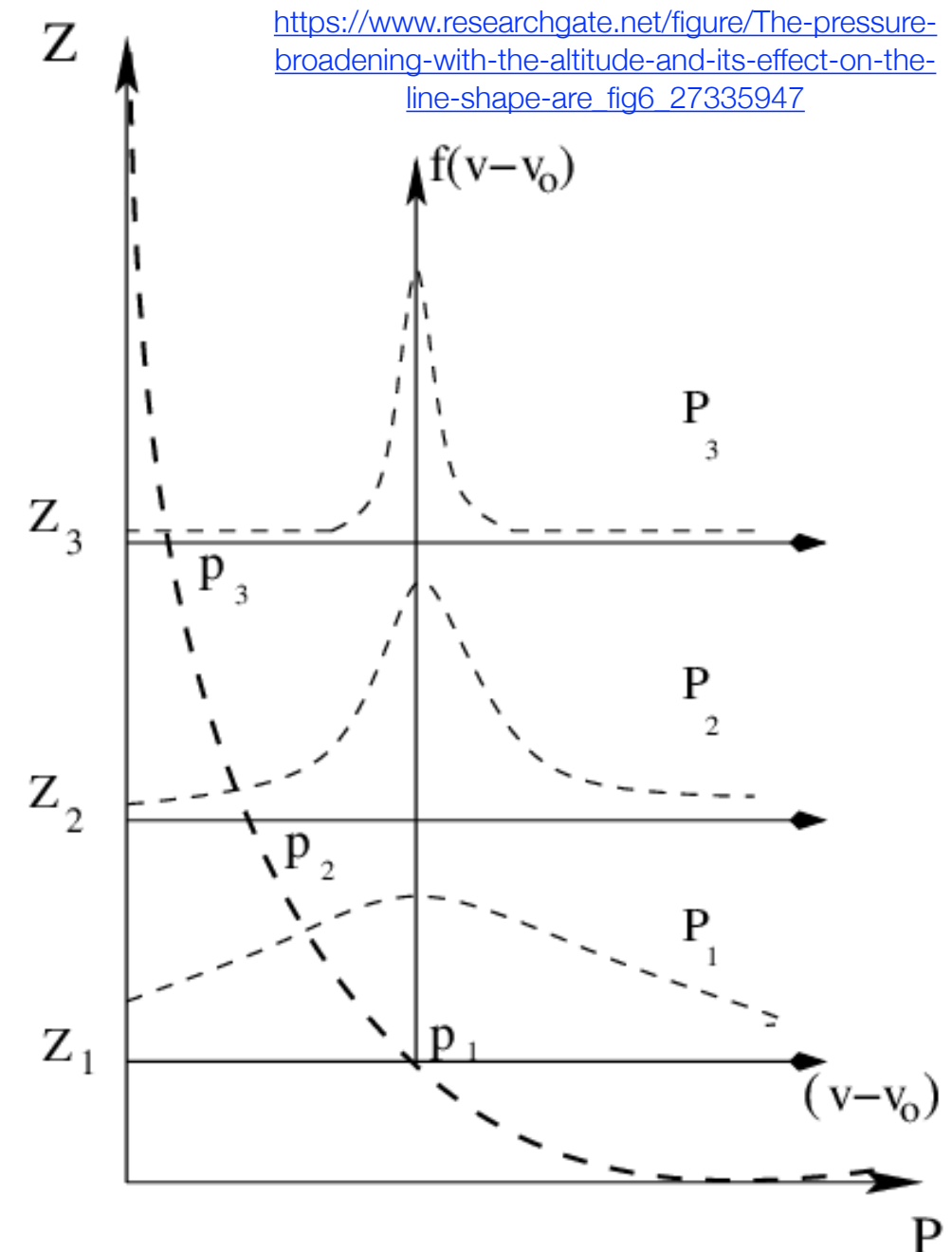
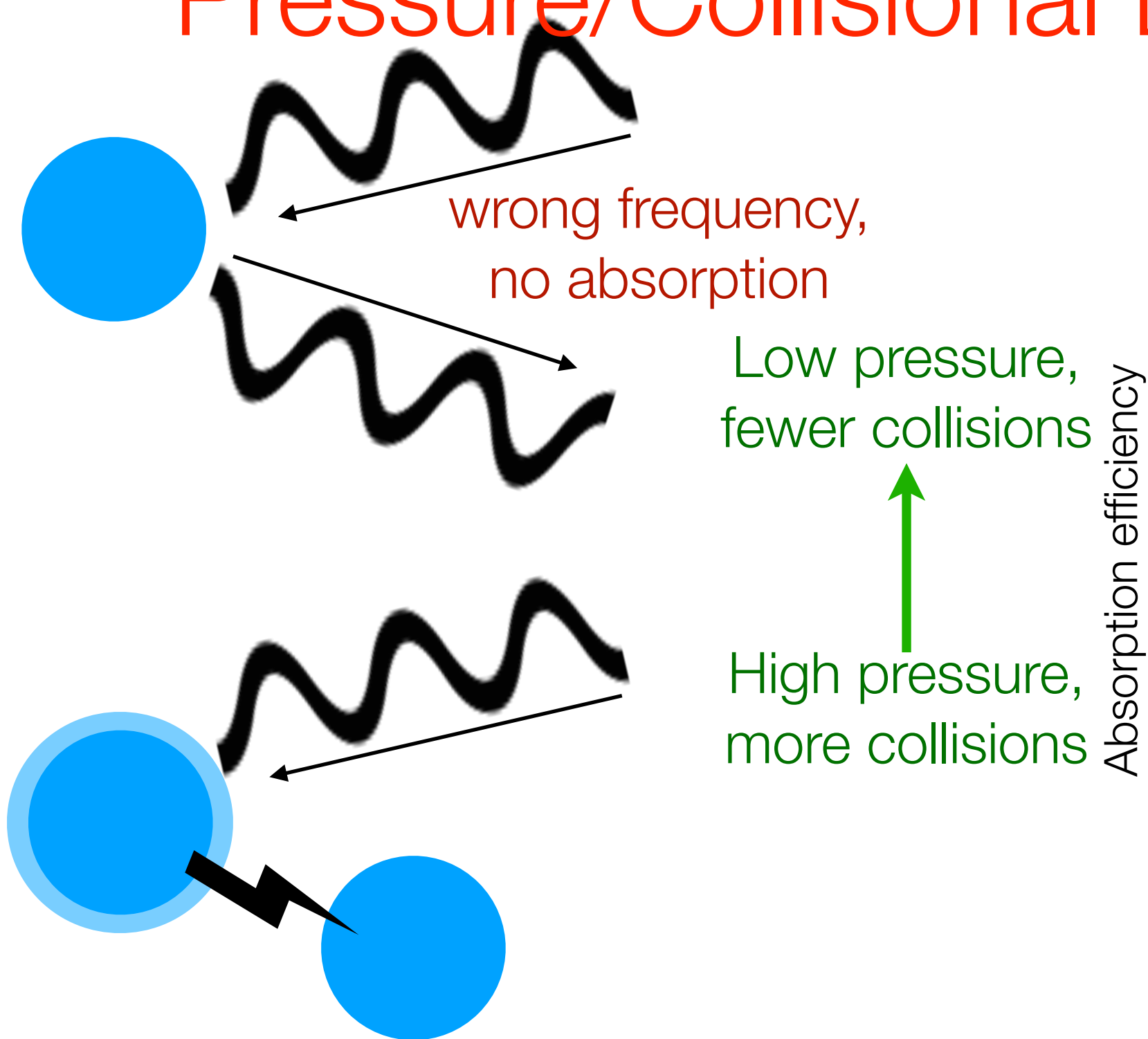
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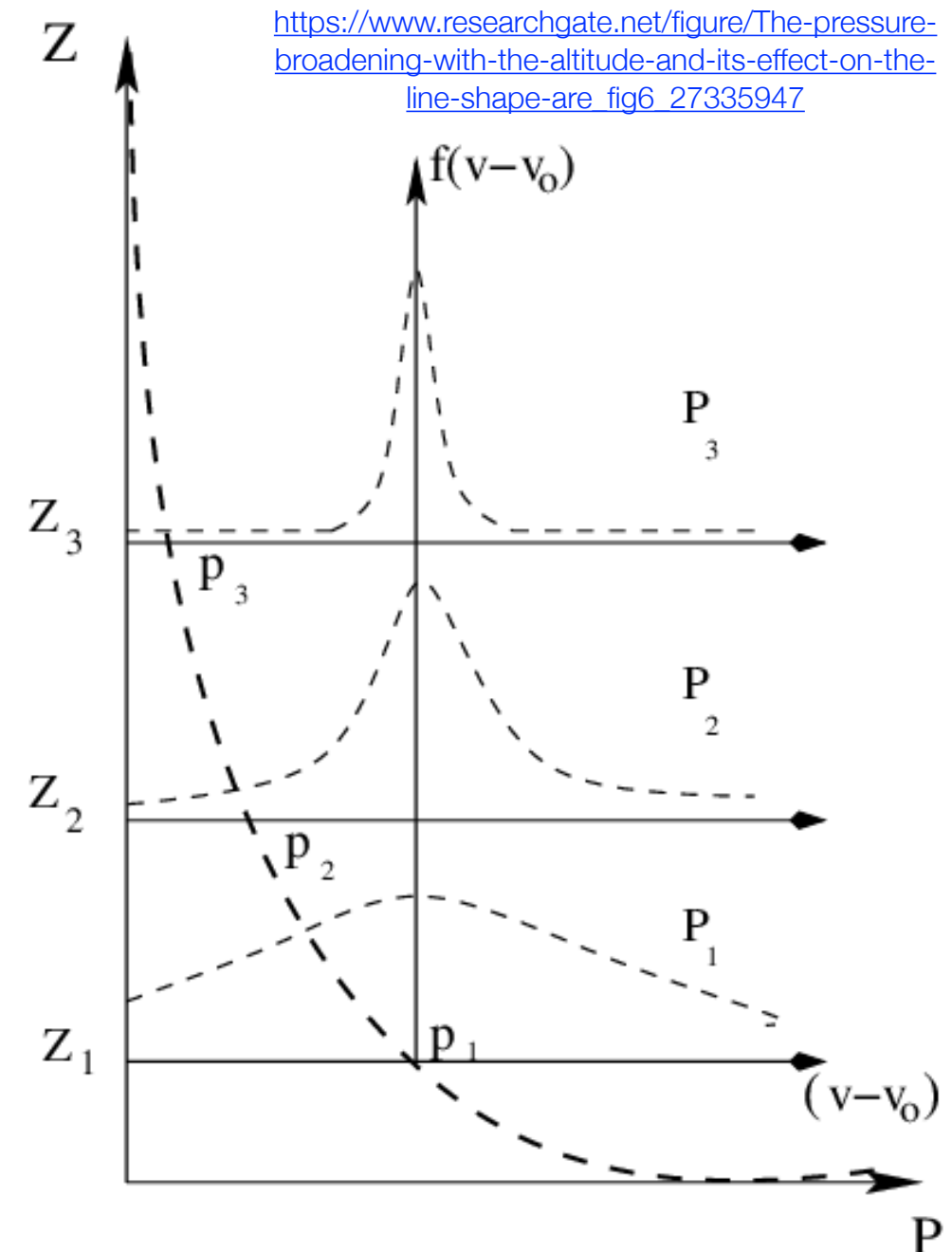
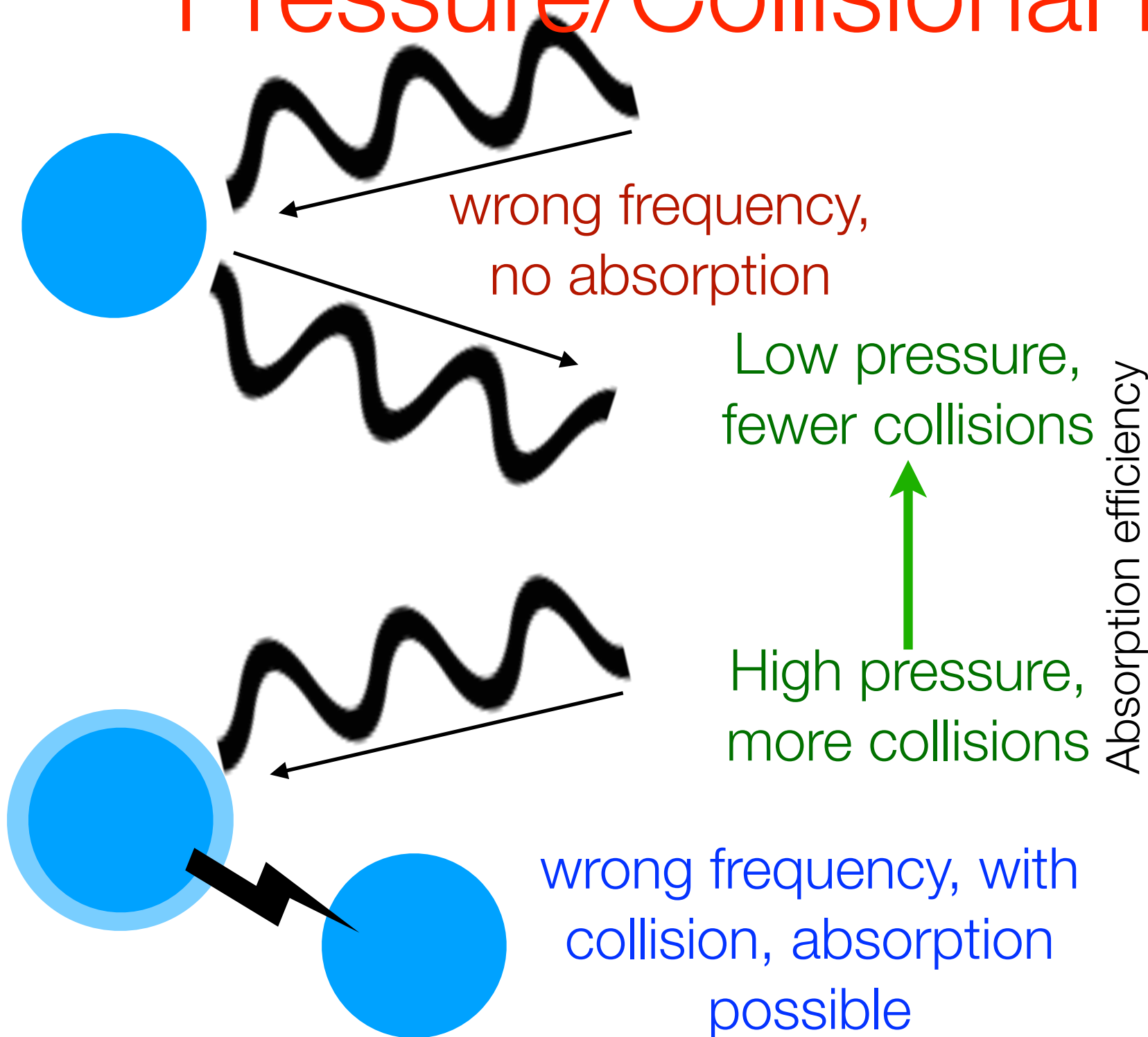
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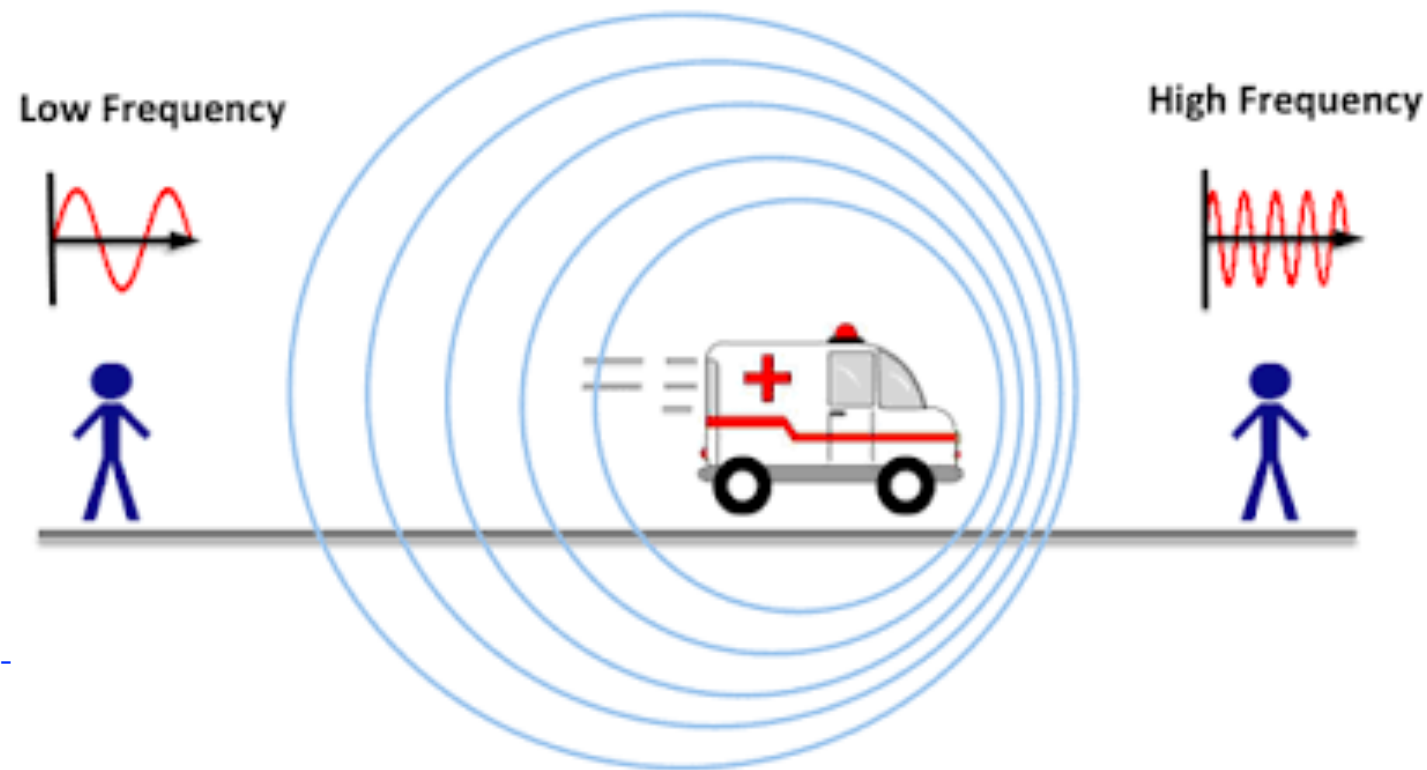


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Doppler Broadening (significantly weaker in Earth's atmosphere)

Doppler Effect



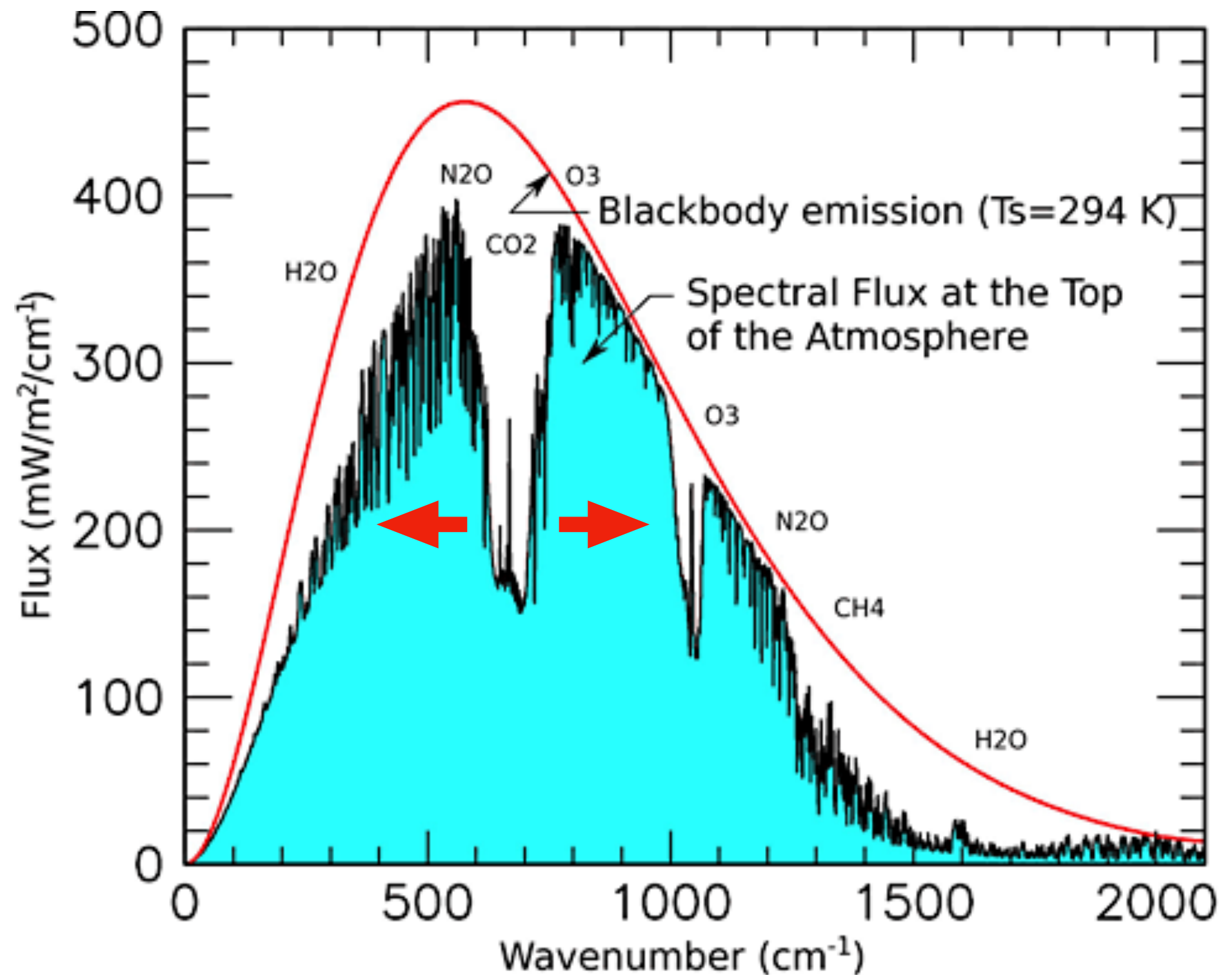
<https://forum.huawei.com/enterprise/en/what-is-the-doppler-effect/thread/510221-100305>

CO₂ molecule moves toward photon ➡ photon seems at higher frequency ➡ molecule absorbs a photon of lower frequency than that of absorption line.

CO₂ molecule moves away from photon ➡ opposite

The random motion of the gas molecules causes the widening of absorption lines because molecules that happen to be moving towards/away from the incoming photon will see it at a different frequency/ wavelengths.

Outgoing Longwave Radiation (OLR)



https://www.giss.nasa.gov/research/briefs/schmidt_05/

note absorption windows...

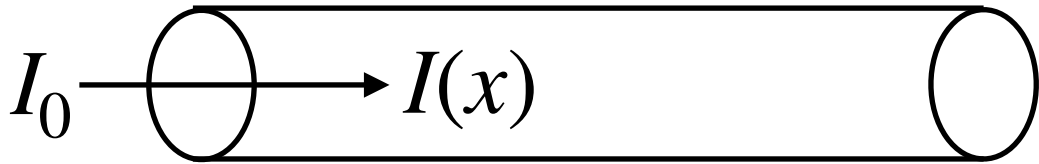
Workshop #3

radiative forcing

Notes sections 2.2.4, 2.2.5:
logarithmic dependence, global warming potential
(use next 5 slides)

Logarithmic dependence of RF on CO₂

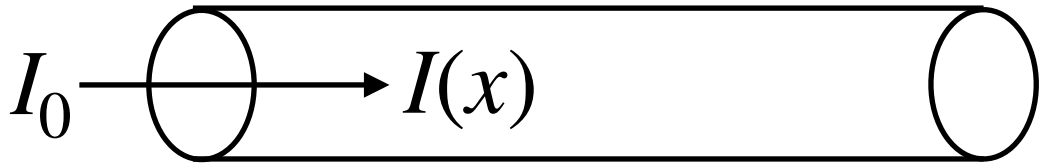
The absorption is typically proportional to the radiation intensity at any given location, and therefore the equation for the transmitted radiation, $I(x)$, may be written as $dI/dx = -\mu I$, where μ depends on the absorbing medium. The solution for the radiation intensity along the pipe is, therefore,

$$I(x) = I_0 e^{-\mu x}$$


The diagram shows a horizontal pipe represented by two parallel lines. At the left end, there is a circle with an arrow pointing into the pipe, labeled I_0 . At the right end, there is a circle with an arrow pointing out of the pipe, labeled $I(x)$.

Logarithmic dependence of RF on CO₂

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For CO₂, the RF varies logarithmically with concentration. We showed in section 3.2 that the warming and radiative forcing are approximately linearly related. This implies that the temperature response also depends logarithmically on the CO₂ concentration. We, therefore, write, schematically,

$$T = T_0 + A \log_2(\text{CO}_2/280)$$

which implies:

$$T_{\times 2} - T_{\times 1} = A \log_2(2\text{CO}_2/280) - A \log_2(\text{CO}_2/280) = A \log_2 2 = A,$$

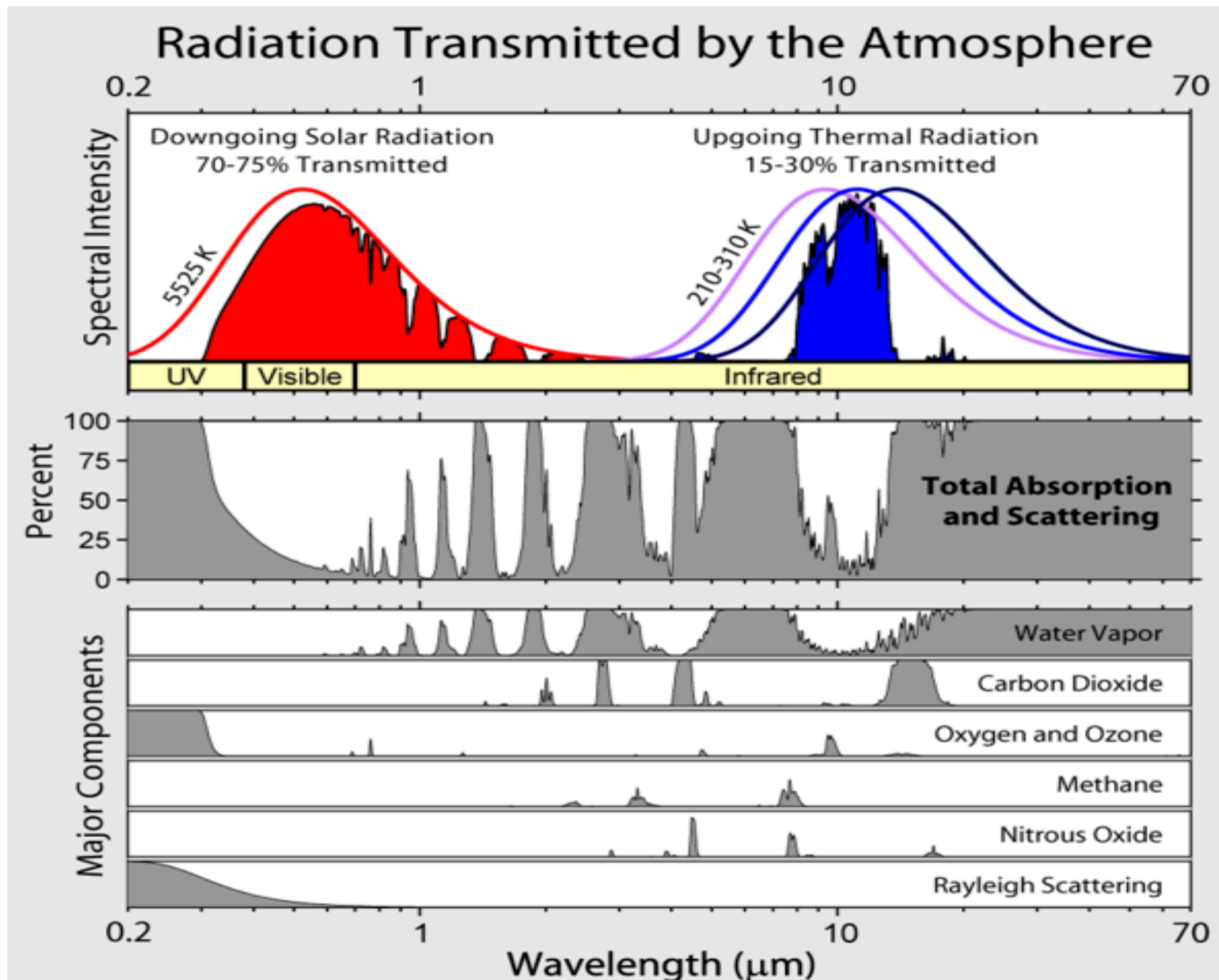
$$T_{\times 4} - T_{\times 2} = A \log_2(4\text{CO}_2/280) - A \log_2(2\text{CO}_2/280) = A \log_2 2 = A.$$

That is, each doubling of CO₂ leads to the same increase in temperature!

What about other greenhouse gases?

How do we quantify their effect relative to that of CO₂?

IR absorption of the major greenhouse gases



https://en.wikipedia.org/wiki/Absorption_band

CO₂ & water vapor absorbs the most IR, at different wavelengths

Other GHG: Global Warming Potential (GWP)

GWP: the time-integrated Radiative Forcing (RF) due to a pulse emission of a GHG, relative to a pulse emission of an equal mass of CO₂

$$GWP(x) = \frac{\int_0^{TH} a_x \cdot [x(t)] dt}{\int_0^{TH} a_{CO_2} \cdot [CO_2(t)] dt} = \frac{AGWP(x)}{AGWP(CO_2)}$$

$x(t)$: temporal decay of a GHG;

$CO_2(t)$: that of CO₂;

a_{CO_2}, a_r : Radiative Forcing per 1 kg (W/m²/kg).

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➡ The GWP of a greenhouse gas is its radiative forcing effect relative to CO₂, given its lifetime and strength as an absorber, over a specified time horizon (TH).

AGWP: absolute GWP.

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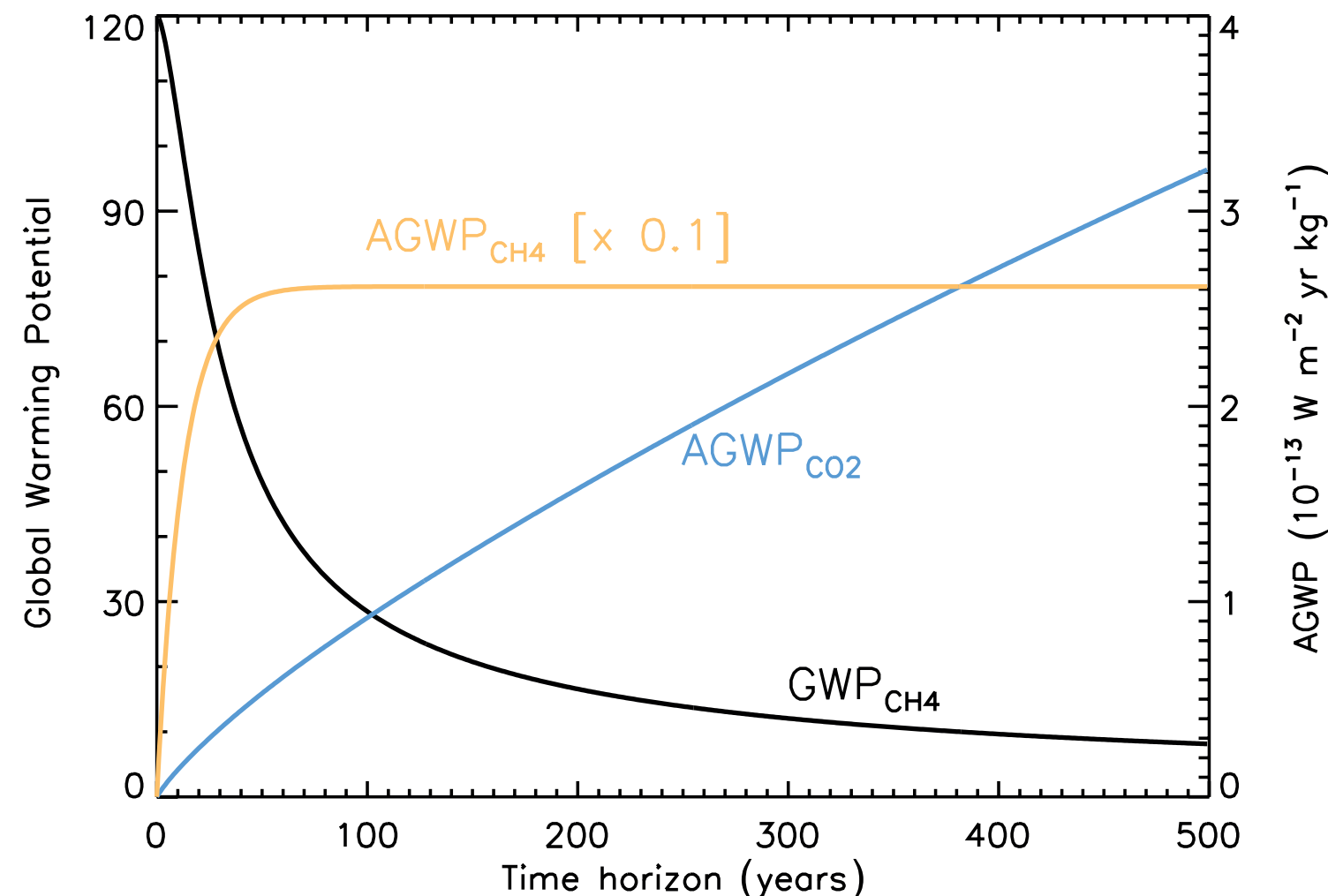
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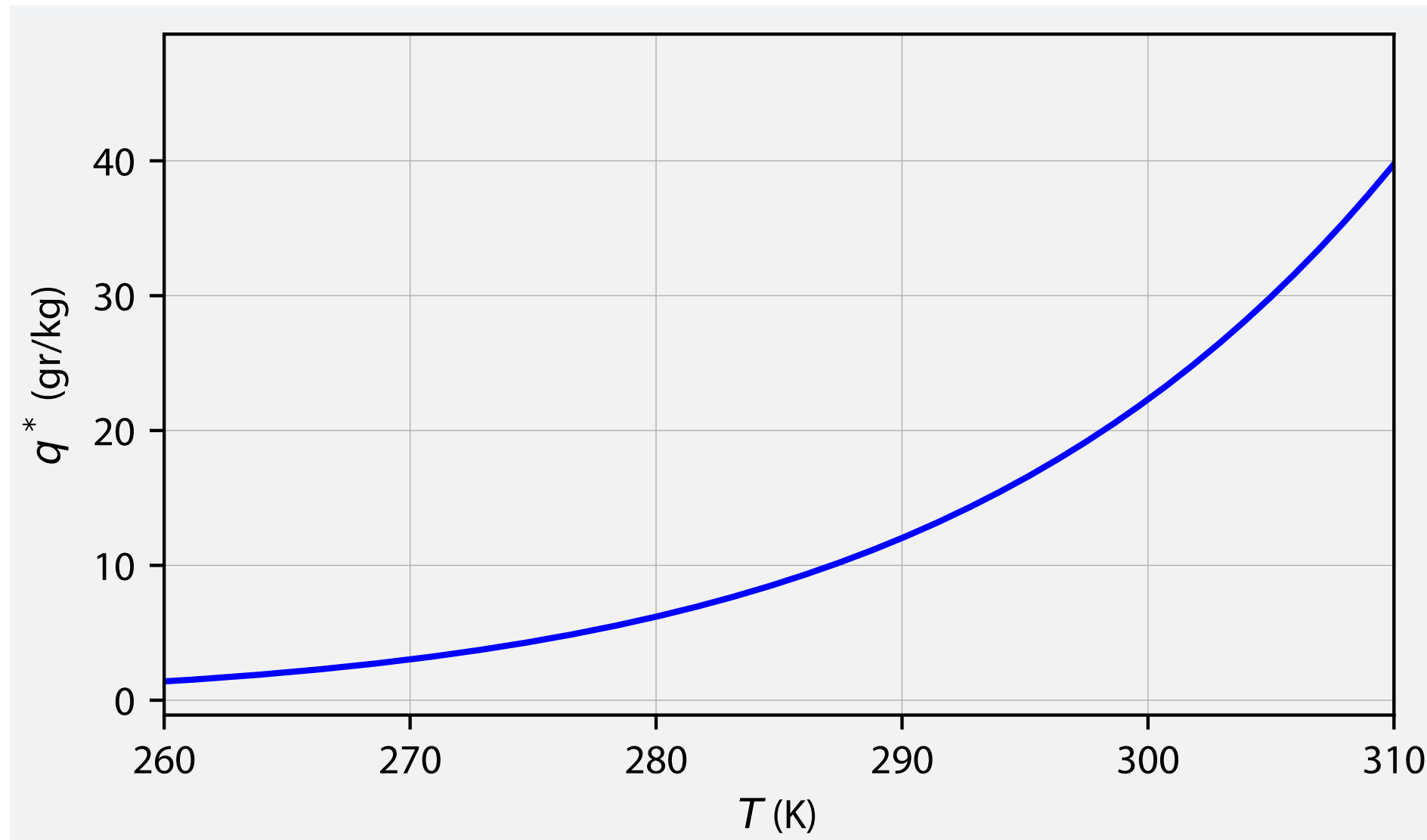
GWP values and lifetimes	Lifetime in years	Global Warming Potential (GWP)		
		20 years	100 years	
Methane	12.4	86	34	
Nitrous oxide (N ₂ O)	121.0	268	298	
Nitrous oxide (N ₂ O)	121.0	264	265	
HFC-134a (hydrofluorocarbon)	13.4	3790	1550	

Notes sections 2.2.6:
water vapor feedback

(use next 2 slides)

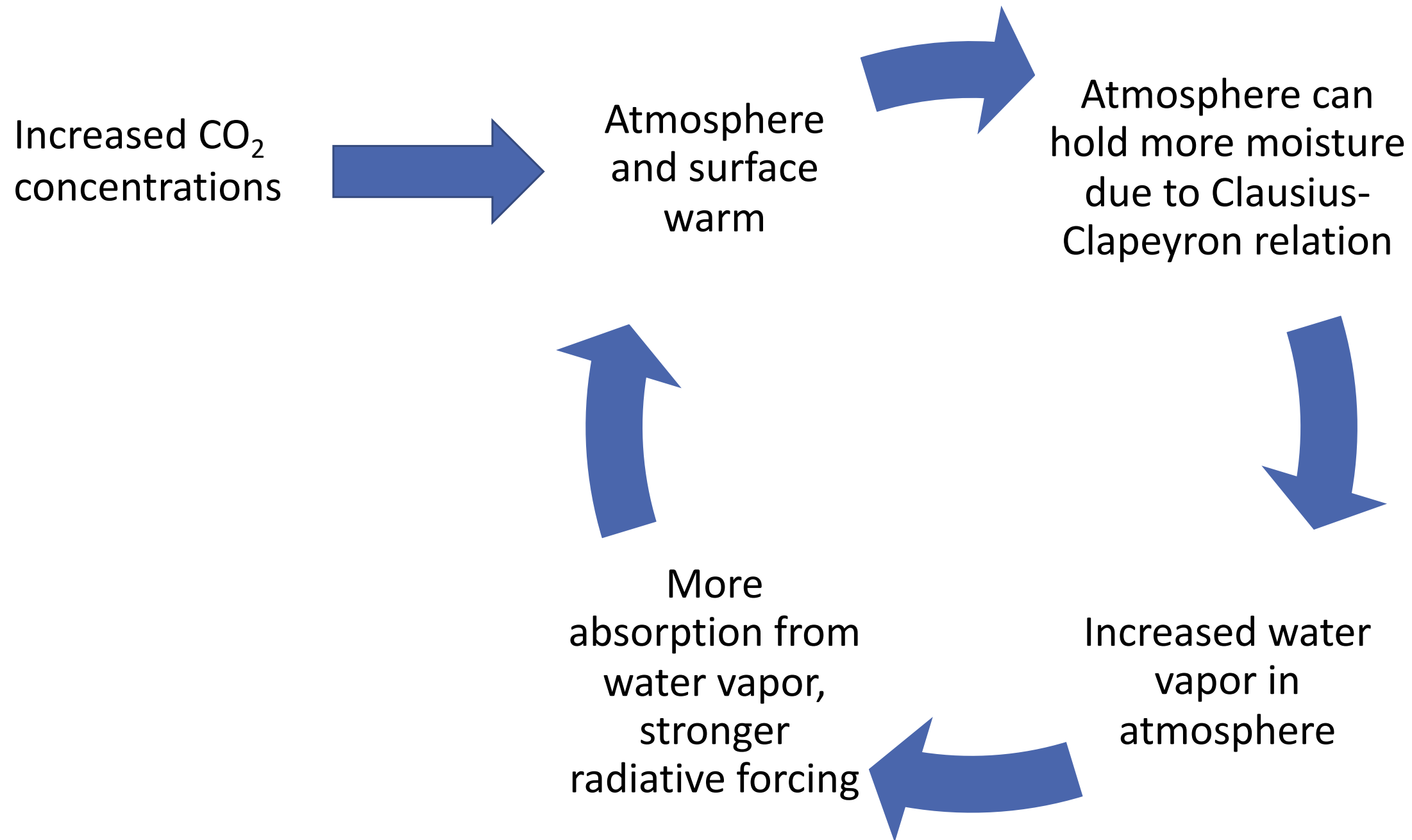
The water vapor feedback

First, a reminder: the Clausius-Clapeyron relation



The saturation water vapor $q^*(T)$ increases exponentially with temperature, at about 6%/ °C

The water vapor feedback



Direct radiative forcing of absorption by water vapor molecules reinforces that by CO₂ via the water vapor feedback

Workshops #4,5: logarithmic dependence global warming potential

Conclusions

- Climate cools/warms to reach radiative equilibrium: incoming SW = outgoing LW

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- The Global warming potential of other GHGs depends on both their efficiency and life time, CO₂ has an especially long life time in the atmosphere

The End